

Title: TO BE FILLED LATER

Authors: TO BE FILLED LATER

Affiliations: TO BE FILLED LATER

Corresponding author email: TO BE FILLED LATER

Abstract

Drinking-water monitoring programmes decide what to measure next, where and with which instrument, often using evidence assembled for other purposes. We developed an evidence-to-action framework that links record harmonisation, validation design, external transfer and documentary appraisal of instruments, and returns for each monitoring context an action with the evidence required to reassess it. Applied retrospectively to 26 contexts in India and in local, not national, archives from Kwale County (Kenya) and Malawi, it returned eight distinct actions; its logic was also applied to published arsenic-kit evidence (Bangladesh) and testing costs (Ghana). At an assumed action level of 0.10 (fraction of records above threshold), further records could change the classification of at least 17 of 160 eligible Indian state-level parameter cells. Across seven tasks in a multinational fitting archive, mean area under the receiver operating characteristic curve was 0.9396 with random record splits but 0.7696 with held-out spatial blocks (one split per design), illustrating why a model's permitted use is tied to how it was validated. Separately fitted models had supported rank transfer in 5 of 13 external archive $\times$ parameter pairs, four in Indian groundwater, permitting sequenced confirmatory sampling within those archives; calibration slopes were below 1 on all 13 and, without local recalibration, no transferred score was used as a probability. In a documentary audit of 23 products, 891 of 1,242 product-criterion cells were unreported in any located document and no product documented all 11 active criteria, so none was named. Because the framework records, for each action, the rule and evidence it rests on, other programmes could apply it to locally verified data; none has yet done so, and whether the actions improve monitoring has not been tested prospectively.

Introduction

Evidence on drinking-water quality still covers fewer people than monitoring is meant to protect. Over the Sustainable Development Goal period, the share of the global population covered by total estimates of safely managed drinking water rose from 35% to 72%, and such estimates became available for India for the first time only in 2025 [1]. Surveillance is also less frequent than required: in urban areas, 21 of 102 countries reported drinking-water surveillance at 95–100% of the required frequency [2]. Infrequent sampling can overestimate how safe water is [3]. A programme with a fixed testing budget therefore has to decide which parameters to measure, at which water points and with which instrument, on evidence that is thin and unevenly spread.

Public archives record where and when testing happened, not the risk across a population. In India, the Central Ground Water Board's current monitoring design samples every quality station once every five years before the monsoon and at least a quarter of the stations before and after the monsoon every year [4]. An exceedance rate computed from records collected under any such design reflects the sampling as much as the water. Measuring where the answer is already settled spends the budget without changing an action; not measuring where the evidence is ambiguous leaves an action resting on that ambiguity.

One response is to model where contamination is likely. Predictive modelling can relate the probability that a contaminant such as groundwater arsenic exceeds a guideline value to soil, climate and terrain predictors, producing maps meant to guide further testing rather than replace it [5]. National models for India have mapped groundwater fluoride from 12,600 measurements [6] and groundwater arsenic, with the hazard maps proposed to inform the prioritisation of testing [7].

How such models are evaluated decides what they can be used for, and practice differs. Environmental records are spatially autocorrelated, so a split that separates rows at random estimates something other than performance in an

unseen region [8], and revisited sites compound the problem. Ignoring dependence between nearby samples biases model assessment towards optimism [9]; the national Indian models were evaluated on random training and test splits [6, 7]. Discrimination and calibration are likewise different properties: a model can order locations well while its stated probabilities are distorted [10]. Estimated risks can be unreliable even when discrimination is good [11], so a model whose ordering holds in a new setting may still misstate probabilities there.

Technology evidence raises the same question at the next step. Technology assessment characterises what instruments can measure and what deploying them demands, from power supply in remote settings to cleaning, signal drift and cost [12]. A programme needs performance near the threshold at which it must act. A 2003 verification of one arsenic field kit, in which one technical and one non-technical operator analysed identical samples, found false-negative rates of 62% and 33% respectively when they read its colour charts, relative to 10 $\mu\text{g/L}$ [13]. A microbial water-quality test has been costed at 21.0 ± 11.3 USD on average across 18 African monitoring institutions in six countries, once consumables, equipment, labour and logistics are counted, with the $\pm$ not defined in the source [14].

Decision analysis makes trade-offs between objectives explicit and can carry uncertainty through to the comparison of alternatives [15]; probabilistic forms report how probable each rank order is [16]. A ranking method returns a leader whether or not the criteria behind it are documented. Machine-learning models always make a prediction unless they are designed to abstain [17]. Neither, by default, reports that the evidence is too thin for the intended use.

Monitoring decisions therefore depend on several bodies of evidence, each of which can be strong enough for one use but not the next. Part of this difficulty is recognised: a review of machine-learning prediction of United States drinking-water contaminants named accounting for spatial autocorrelation in model training among the remaining methodological challenges [18]. We developed an evidence-to-action framework that connects record harmonisation, validation design, external transfer and documentary appraisal of instruments. At each step it tests whether the evidence supports the use a programme would make of it, and for each monitoring context it returns an action together with the evidence required to reassess that action.

We applied the framework retrospectively to archived and published evidence: India as the principal application, local archives from Kwale County in Kenya and from Malawi, and decisions in Bangladesh and Ghana that rest on published evidence alone. The applications address four questions a programme faces: where further records could change a monitoring classification; how much validation design changes the performance estimate it would rely on, and whether a model's ordering and probabilities carry over to groundwater archives that took no part in fitting; what records alone, or published evidence alone, support where no model is used; and whether the documentation of named instruments is sufficient to compare them. Retrospectively, the framework returned an action and the evidence required to reassess it for every context, separated rank use from probability use where models transferred, and named no product because the located documentation did not support a comparison; whether the actions improve monitoring was not tested.

Materials and methods

Evidence-to-action framework

The framework takes three bodies of evidence (an accepted monitoring archive, the output of any predictive model, and the documentary evidence about candidate instruments) and returns two things for each monitoring context, a country $\times$ source archive $\times$ parameter combination (Fig 1). The first is an action, which names what a programme should do next rather than scoring a place. Actions run from acquiring a baseline or repairing unusable records, through prioritising confirmatory sampling, to direct or targeted reference measurement and sentinel monitoring paired with chemistry; S2 Table defines all eight. The second return is the evidence required to reassess the action: the additional evidence that could move the context to a different action, stated as a kind of evidence rather than as a sample size, cost or optimal design. Three gates stand between the inputs and the output: the validation gate at step 6 of the workflow, the permission gate at step 7 and the evidence gate at step 8.

The validation gate settles in which role a model may enter a decision: ordering locations so that confirmatory sampling is sequenced, supplying a probability to compare with an action level, or neither. The role follows from the validation design, not from the size of a reported metric. Monitoring records are not exchangeable rows, because sites are revisited and nearby observations share geology, land cover and often a sampling programme, so a random split mostly measures interpolation within the existing sampling structure. Keeping every record from a site together removes the repeat-visit component; holding out contiguous spatial blocks additionally requires the model to

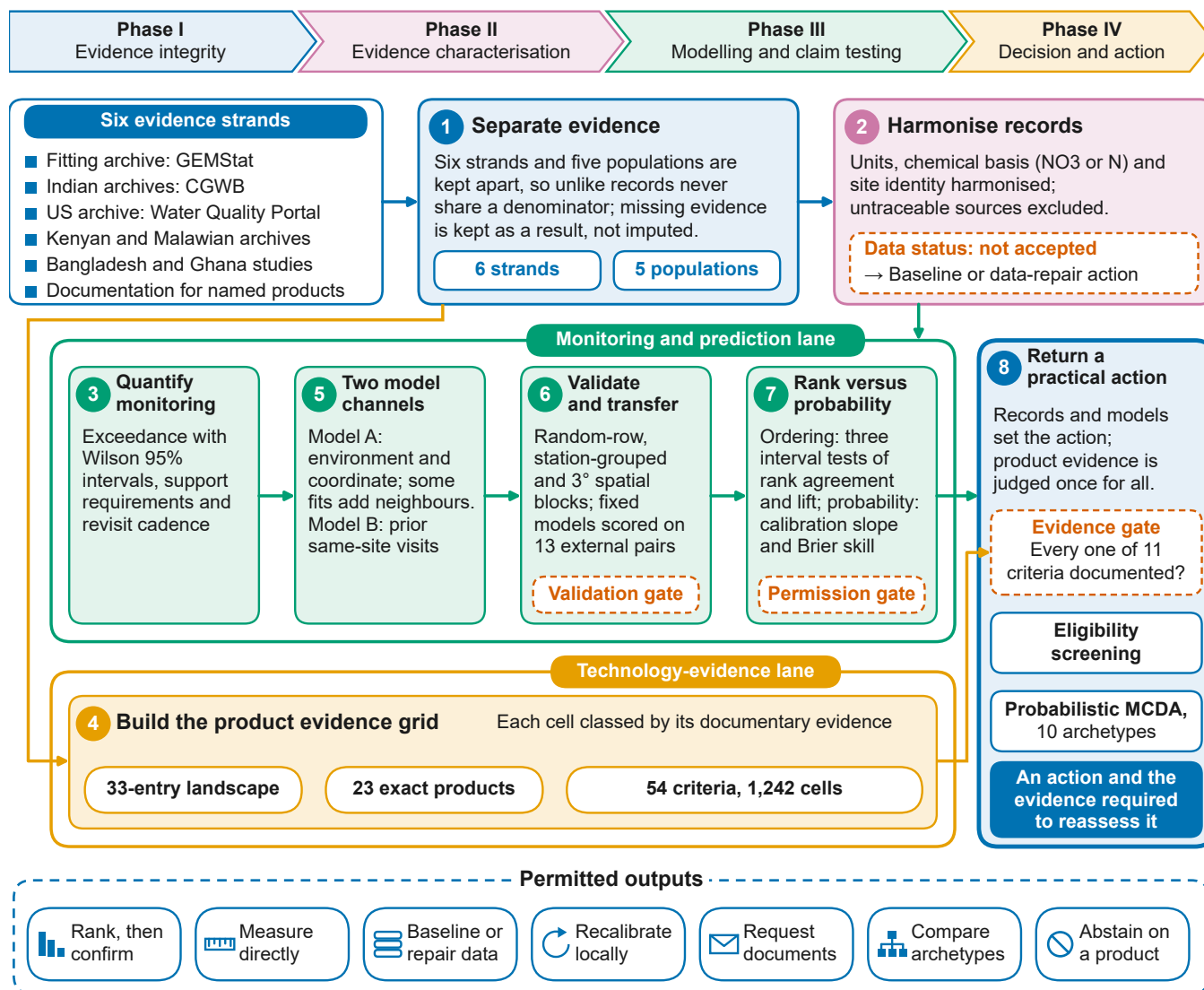


Fig 1. Evidence-to-action framework. Eight steps connect separate evidence strands to monitoring actions through three gates: validation (step 6), permission to use ordering or probability (step 7) and product documentation (step 8). Each output is a kind of action, stated with the evidence required to reassess it. MCDA, multi-criteria decision analysis; CGWB, Central Ground Water Board.

generalise across geography [8]. The gate prevents a random-row figure from being read as performance in a district that contributed no training station.

Discrimination asks whether the model orders cases correctly; calibration asks whether a stated probability matches the frequency actually observed at that probability [10]. Calibration here means model calibration, not the calibration of an instrument. A ranking tells a programme where to look first, and a probability compared with a threshold tells it when to act, so the permission gate recognises three states. Where ordering survives a move to a new setting but calibration there has not been shown to be adequate, the model may set the order of confirmatory measurement. Where calibration on local labels (direct local measurements used to check a model's scores) has also been assessed and found adequate, or restored by local recalibration, a probability may be compared with a fixed action level. Where ordering does not survive, direct measurement is required. The gate is applied to each archive $\times$ parameter pair, because two archives holding the same analyte can return opposite verdicts.

Once a decision requires direct measurement it becomes a question about an instrument, governed by the evidence gate through two rules. The documentation threshold asks whether a criterion is documented for enough products for a comparison on it to be informative. The sufficiency rule asks whether a given product documents every active criterion, the 11 criteria a comparison requires (S1 Text), and passes its must-pass analytical, matrix, deployment and identity requirements. A product that fails the sufficiency rule is not scored: scoring an undocumented field at a middling value would rank a device that reports a poor maintenance interval below one that reports nothing. Missing documentation is unknown evidence about a product, not evidence that it performs poorly. Where no product passes the sufficiency rule, the framework abstains at product level and states the documents that would lift the abstention. Multi-criteria comparison of technology archetypes gives preference context for a class of instrument under stated weights without establishing anything about a product within that class.

Abstention is accordingly a designed output with the same role as the reject option of a selective classifier [19], but triggered by missing evidence rather than by expected error (see Discussion), and not a failure. The record and model gates are met in order: a context whose records cannot be accepted returns a baseline or repair action before any model is considered, and one whose model output fails the validation or permission gate returns direct measurement. The evidence gate is applied once, to the product catalogue, and its determination holds for every context before any instrument is named. The rules are the same whatever a product documents, so no product's values can alter the gates it is judged by.

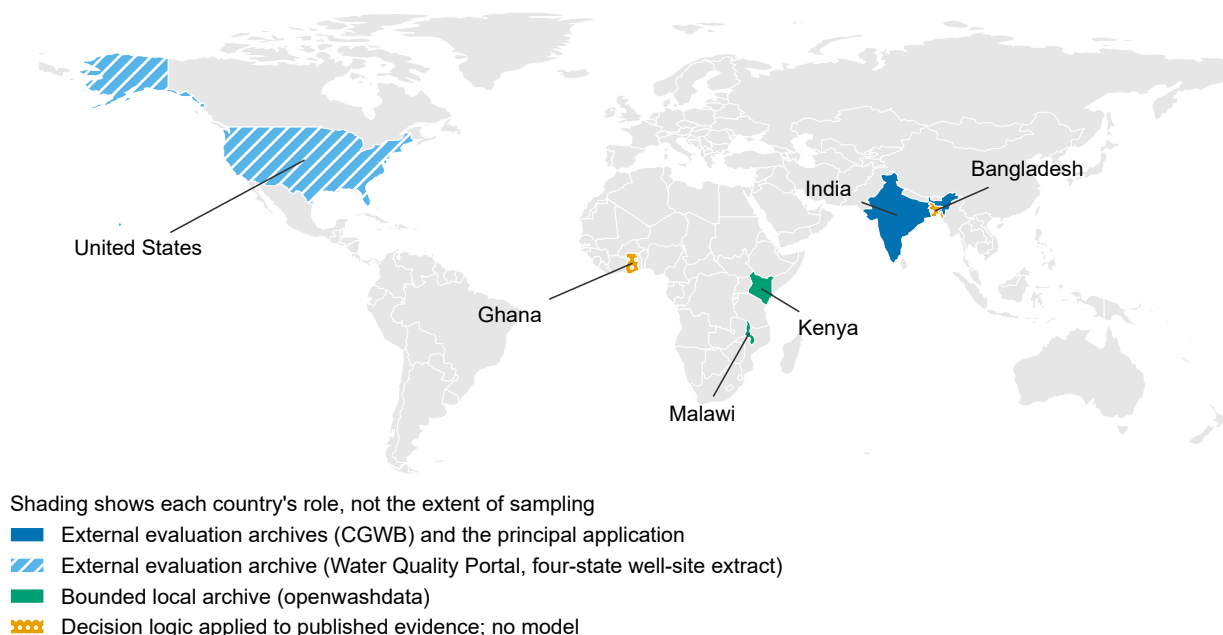
The framework separates three monitoring functions, because devices compared before the function is fixed were never doing the same job: distributed spot screening of existing water points; a higher-frequency sentinel at selected sites, which buys temporal resolution at the cost of power, logging, drift, fouling and downtime; and confirmatory or reference measurement of a specific contaminant. The functions combine within one monitoring architecture, and each setting's action names the function it relies on.

Study design and unit of analysis

The unit of analysis for decisions is the monitoring context, one country $\times$ source archive $\times$ parameter combination, and 26 were analysed, 8 in India, 8 in Kenya and 10 in Malawi (generation rules in S1 Text). The Kenyan and Malawian archives are bounded local archives, not national samples, and no result from them is a national estimate for either country. Bangladesh and Ghana are applications of the decision logic to published evidence, with no model fitted or transferred; their studies were chosen for the decision each setting poses rather than found by a systematic search, so other studies could change the evidence those applications rest on. Five populations are kept separate and never merged: 23 exact commercial products, a 33-entry technology landscape (Fig 1), reference laboratory methods, emerging research classes, and ten technology archetypes, classes of instrument compared by multi-criteria decision analysis (MCDA). Missing evidence was retained, not imputed.

Data sources and evidence roles

Six evidence strands enter the analysis (Fig 2): the fitting archive, the Indian archives, the United States archive, the Kenyan and Malawian archives, published studies from Bangladesh and Ghana, and documentation for named products. The fitting archive is the GEMStat global water-quality database of the UNEP GEMS/Water Programme [20]: a multinational archive, predominantly of surface water, used only for model fitting and internal evaluation. Seven classification tasks were built from it, for arsenic, nitrate, conductivity, suspended solids, turbidity, fluoride and pH (thresholds in S1 Text).



*Fitting archive (GEMStat): multinational and predominantly surface water.
Station locations are not drawn.*

Fig 2. Evidence roles by country. Shading shows the role of each country's evidence, not the extent of sampling: external evaluation archives in India and a four-state United States well-site extract, bounded local archives in Kenya and Malawi, and published evidence only for Bangladesh and Ghana. Boundaries: Natural Earth. Data: S9 Data.

Two external arms, neither of which took part in fitting, were used to evaluate transfer. The primary arm is Indian groundwater from the Central Ground Water Board (CGWB), in two products that share a producer: National Water Data Portal records for pH, conductivity, total dissolved solids (TDS) and arsenic (the CGWB portal), and fluoride and nitrate tables recovered from the CGWB Ground Water Quality Data documents (the CGWB Year Book) [21–24]. CGWB samples groundwater from observation and monitoring wells, with the stated aim of assessing shallow aquifers against the Indian drinking-water standard [25], so a record describes groundwater at a monitoring well, not water as supplied. The second arm, WQP, is a 2023 well-site extract for four states from the United States Water Quality Portal [26] (S1 Text). Both external arms are groundwater, so transfer from the fitting archive crosses water-body type as well as geography. Independence rests on archive lineage and a 0.1° cell exclusion between training and external locations, not on a record-level audit, so the external archives are not claimed to be fully independent of the fitting archive.

The bounded local archives are three openwashdata packages published under CC BY: samples from two 2016 campaigns in Kwale County, Kenya [27]; a laboratory snapshot of 32 Malawian water points, 30 of them boreholes or tubewells, sampled in three rounds with different parameter suites [28]; and a Malawian conductivity series from loggers in monitoring wells [29]. Surface-water samples were excluded from the Kwale archive, which records no water-point type. Acceptance checked units, identity and plausibility; no charge-balance or conductivity–TDS consistency check was applied, and no records from other Indian or Kenyan custodians, microbial records included, were used. In the Indian archive, arsenic values labelled mg/L but implausible as such, from four north-eastern states in 2017, were divided by 1,000 (S1 Text). Harmonisation rules, versions and accepted record counts are in S1 Text.

Thresholds and chemical basis

An exceedance rate is a record fraction, the proportion of accepted records above a threshold, reported with its numerator and denominator. Fluoride above 1.5 mg/L and arsenic above 0.010 mg/L follow the WHO guideline values, the arsenic value being provisional [30]. These WHO values are applied to every archive, whereas CGWB judges Indian drinking suitability against the Indian standard IS 10500 [25], whose limits S1 Text compares with ours. Nitrate was classified above 50 mg/L as NO₃, the WHO guideline value expressed as nitrate ion [30]. The transfer tasks use nitrate above 10 mg/L as N, the value of the United States maximum contaminant level [31]. That is 44.268

mg/L as NO_3 , so no nitrate figure is carried between the two analyses. WHO proposes no health-based guideline value for pH or TDS: 6.5–8.5 is the usual optimum pH range for supply, and water becomes increasingly unpalatable at TDS above about 1,000 mg/L [30]; both are used here as operational ranges. Conductivity above 1,500 $\mu\text{S}/\text{cm}$ is an unverified operational project convention, not a health guideline or an Indian drinking-water limit; under the factor CGWB recommends, TDS of about $0.65 \times \text{conductivity}$ [4], it corresponds to about 975 mg/L TDS. The 1 NTU turbidity rule is WHO’s value for supporting effective disinfection [30], applied here to untreated water as a project rule. Microbial results use a 100 mL basis, and faecal coliform is never relabelled as *E. coli*.

Binomial uncertainty on record-level exceedance proportions is described by Wilson 95% intervals [32], which do not remove spatial clustering or sampling-design bias, so rates are reported as monitoring-record signals, not as population prevalence or exposure.

State-level cells and local archive summaries

For India we formed state or Union Territory $\times$ parameter cells from the accepted CGWB records. A cell was described when it held at least 30 records at 10 or more sites; its median revisit interval, on a site key built from the station or well identifier and the sample date, and its support for leave-one-state-out and temporal analysis were recorded (S3 Table). A cell is decision-relevant when the Wilson 95% interval on its exceedance rate straddles an action level, set at 0.10 as a stated assumption and not a regulatory value, with counts also reported at 0.05 and 0.25. This classification allocates confirmatory effort between states and parameters and sets no action; CGWB’s own response to an exceedance is at station level, monitoring affected areas as trend wells and new hotspots on a 2 km grid [4]. For Kwale and the Malawi laboratory snapshot we report record fractions against the same thresholds. For the 67 Kwale sites sampled in both campaigns (66 for nitrate and fluoride) we compared threshold states between campaigns using site medians, and for the Malawi logger series we counted changes of state between successive daily site medians.

Models and validation designs

Two information channels were modelled. Model A uses environmental and terrain covariates and the site coordinate, with nearest-station neighbour features and month added in the transfer and fold-comparison fits (S1 Text); Model B uses earlier measurements at the same site and needs at least two of them. Both use the histogram-based gradient-boosting classifier in scikit-learn [33], with configurations fixed in advance and no hyperparameter search. Model A’s configuration is in S1 Text; Model B’s was not retained.

Three validation designs were compared on the seven fitting tasks (Fig 3). Each split the fitting archive once into training, validation and test partitions keyed on the record (random-row), the station (station-grouped) or a contiguous 3° spatial block; Model A was fitted on 24 environmental, terrain and coordinate features, calibrated isotonically on the validation partition and scored once on the test partition (S1 Text). Block splitting keeps nearby observations together and is the design used here to address generalisation to held-out blocks within the sampled geography [8]; no autocorrelation range was estimated to set the block size. ROC-AUC (area under the receiver operating characteristic curve), precision–recall AUC, sensitivity, specificity, calibration slope and Brier skill were interpreted separately; precision–recall AUC is read against prevalence as its no-skill baseline [34], and a 0.5 cut-point was used diagnostically rather than as an operational default. Prevalence here is a task’s base rate, the share of its records above threshold, not population prevalence. A calibration slope between 0 and 1 means predicted probabilities are too extreme, and a slope at or below 0 means they carry no usable probability ordering. The two channels were compared separately, in forward-chained yearly folds on records with at least two earlier same-site measurements, with DeLong tests on the pooled predictions (S1 Text). No local recalibration was performed, because it needs held-out predictions that were not retained.

A mechanistic ablation tested whether aquifer, soil, land-cover and population variables substitute for raw coordinates for conductivity and pH; its result, whose intervals overlap, is in S5 Text, and S6 Fig shows one illustrative surface, read as an ordering and not as a probability. Its stack comprises the national aquifer classification [35], SoilGrids 2.0 soil properties [36], ESA WorldCover 10 m 2021 v200 land cover [37] and WorldPop population density [38].

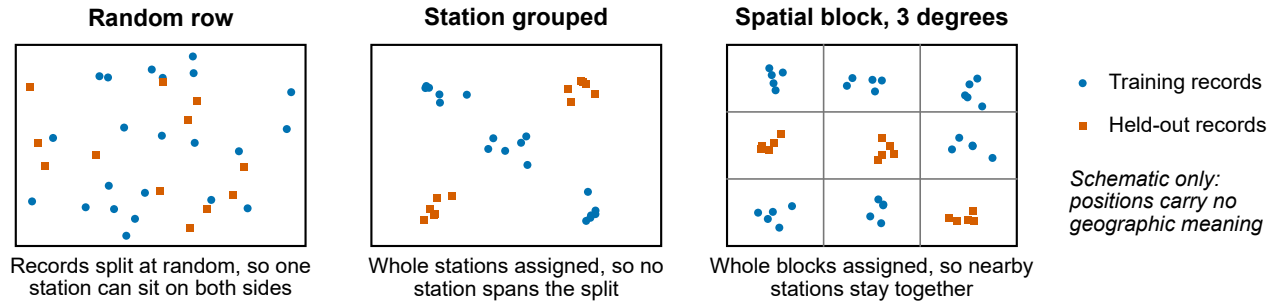


Fig 3. Three validation designs. How each design splits monitoring records into training and held-out sets: by random rows, by whole stations or by contiguous 3° spatial blocks, with one partition per design. Positions are schematic and carry no geographic meaning.

External transfer assessment

Transfer was assessed per archive $\times$ parameter pair. For each parameter a separate Model A fit was trained on GEMStat records before a cut year, on 33 features that add nearest-station neighbour features and month to the 24 above (S1 Text). These transfer fits are not the seven validation fits and were not evaluated under the three designs, so the optimism measured under the designs is not carried to the transferred scores. The fixed fits were scored on the external archives with no tuning, threshold selection or recalibration after the external data were seen, once every external site sharing a 0.1° cell with a training station had been discarded. Of 24 possible pairs, 13 were evaluated and 11 are reported as untested rather than negative.

The verdict comes from three interval tests rather than from a threshold on discrimination, each passed when a 95% bootstrap interval lies wholly above its null value: (i) the site-level rank correlation between predicted risk and observed severity, where severity is $\log_{10}$ concentration or, for pH, the distance from the centre of 6.5–8.5 (site bootstrap; above 0); (ii) the rank correlation within 1° blocks (block bootstrap; above 0); and (iii) lift in the top predicted decile (site bootstrap; above 1). With fewer than two tests evaluable a pair has *insufficient support*, a gap in the evidence that is never merged with a negative result; if every evaluable test passes, *rank transfer is supported*; if none passes, it is *not supported*; otherwise it is *partial or scale-dependent*. Only the outcome of each test was retained, not its statistic or interval, so a verdict carries no stated sampling uncertainty. For every pair we also recorded, each with a site-clustered bootstrap interval, threshold discrimination (threshold ROC-AUC) and a detected-record severity correlation (the Spearman rank correlation between predicted risk and observed severity among detected records only, not a verdict test), together with the calibration slope, Brier skill against a reference forecast equal to the external base rate, and confusion counts at a 0.5 cut-point (S1 Text). The retained calibration intercept was estimated jointly with the slope and so is not calibration-in-the-large, and no criterion for adequate calibration was fixed in advance. A supported verdict authorises prioritised confirmatory sampling only, at the level of observations within the archive.

The Kenyan and Malawian archive summaries were recomputed from the raw archives for this article. The Indian state-level results are the outputs of the original analysis, checked for internal consistency but not re-run, and the validation, transfer and ablation results were checked against retained result exports rather than re-run for this article (S1 Text).

Technology evidence and action rules

The exact-product catalogue comprises 23 named commercial products assessed against 54 criteria, a 1,242-cell grid in which each cell records its value, source, locus and evidence class (S4 Table). Not reported, not researched, conflicting and range-only evidence are kept distinct, and evidence about an adjacent product cannot verify a named model. No record states how the 23 products were chosen or which sources were searched and when, so the product counts describe the documentary record as located. A criterion meets the documentation threshold when it is documented for at least 12 of the 23 products, counted on its primary database field (strict) or on any database field mapped to it (any admissible). The sufficiency rule requires all 11 active criteria (S1 Text) to be documented; threshold-near classification error, invalid-test rate and testing frequency are not among the active criteria.

Conditional comparison of the ten technology archetypes used the same 11 criteria under seven declared weighting scenarios and 13 cost mappings, with 40,000 Monte Carlo draws per scenario from a Dirichlet distribution centred on the scenario's weights (S1 Text, S7 Table). The technique for order of preference by similarity to ideal solution

(TOPSIS) supplied the primary rank [39], and stochastic multicriteria acceptability analysis (SMAA) represented weight uncertainty as a distribution over the feasible weights [40]. The comparison is not screened for the analytes a scenario targets, and no practitioner weights were elicited.

For each monitoring context a rule returns an action (S2 Table): for Kenya and Malawi from the data status alone, whatever the exceedance count, and for India from a per-parameter evidence state that the authors assigned from the transfer verdict, record support and identity limits (S1 Text gives both branches). The evidence states and the rule were recorded on 13 August 2026, after the transfer verdicts had been computed, and no dated protocol records when the action level, support requirement, verdict rule, documentation threshold or active criteria were fixed. The product-level decision is a single determination over the 23-product catalogue, applied to all 26 contexts, not 26 independent determinations.

Use of artificial-intelligence tools

OpenAI ChatGPT (version not recorded) was used to set up the project's initial prompts, rules and scripts. Anthropic Claude Opus 5, through Claude Code, was used to write and run the data-processing, verification, table- and figure-building scripts; to search for and select literature, open cited sources, record the passage supporting each citation and assemble the reference list; to draft and revise every section of the text and the supporting information; and to run AI reviews of the drafts, whose accepted recommendations were applied. The model runs behind the validation and transfer results were not produced with these tools, and no tool altered or imputed a measurement. Table and figure values were re-derived by script from their source files, every number in the text was checked against a register of sourced values, and each cited sentence was checked against the passage recorded from its opened source. S10 Text lists each activity, its check and whether the author has reviewed it; at the time of writing, the author's review of the text and of the citation records was outstanding.

Results

We applied the framework first to India, where the accepted archives are largest, then to the local archives of Kwale County in Kenya and of Malawi, and finally to published evidence from Bangladesh and Ghana. Across the 26 monitoring contexts the rule returned eight distinct actions, each with the evidence required to reassess it (Fig 4, Table 1). The most frequent were *local signal plus direct confirmation*, in 9 contexts, and *monitoring baseline required*, in 5; six contexts hold no records, and their actions follow from that absence.

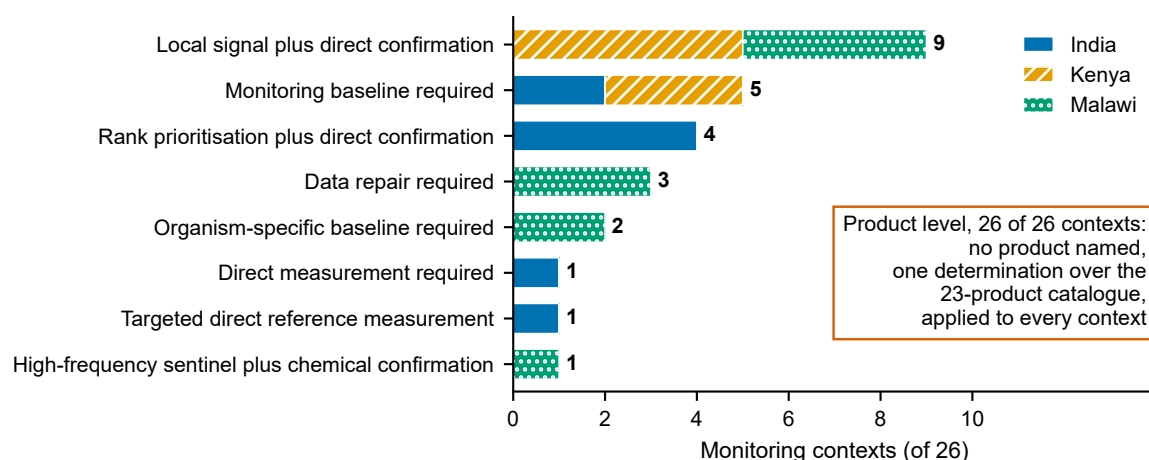


Fig 4. Actions returned in the 26 monitoring contexts. Number of contexts, each a country × source archive × parameter combination, returning each of the eight actions, by country (India 8, Kenya 8, Malawi 10). Data: S9 Data.

Table 1. Monitoring actions and evidence needed for reassessment. One row per country $\times$ source archive $\times$ parameter combination; the full matrix is S2 Table.

Monitoring context	Records (n)	Action	Evidence required to reassess the action
India, CGWB Year Book: fluoride	35,843	Rank prioritisation plus direct confirmation	Repair longitudinal well identity and run same-sample field-versus-reference validation.
India, CGWB Year Book: nitrate	34,642	Rank prioritisation plus direct confirmation	Repair persistent well/date/sample identity and collect repeat measurements before any temporal model.
India, CGWB portal: arsenic	9,927	Targeted direct reference measurement	Recover method and detection-limit metadata and build a defensible detected/non-detected analytical population.
India, CGWB portal: conductivity	173,029	Rank prioritisation plus direct confirmation	Governance review of the 1,500 $\mu\text{S}/\text{cm}$ project convention plus prospective sentinel-versus-spot evaluation; major-ion chemistry at flagged wells.
India, CGWB portal: pH	162,729	Direct measurement required	Collect representative local labels at sentinel sites and assess whether a separate local recalibration or model is useful; pH departures reported by direction.
India, CGWB portal: TDS	30,151	Rank prioritisation plus direct confirmation	Prospective paired conductivity/TDS sampling across relevant water matrices and seasons; recovery of how the portal TDS values were derived.
India, no accepted archive: <i>E. coli</i>	–	Monitoring baseline required	Acquire traceable India microbial measurements with household/source context and compatible denominators.
India, no accepted archive: turbidity	–	Monitoring baseline required	Acquire a machine-readable India archive with site, date, method, unit and valid turbidity results.
Kenya, Kwale groundwater samples: arsenic	140	Local signal plus direct confirmation	Repeat same-site arsenic measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: conductivity	140	Local signal plus direct confirmation	Repeat same-site conductivity measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: <i>E. coli</i>	–	Monitoring baseline required	Acquire traceable local <i>E. coli</i> records with site, date, unit, method and quality-control fields.
Kenya, Kwale groundwater samples: fluoride	139	Local signal plus direct confirmation	Repeat same-site fluoride measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: nitrate	139	Local signal plus direct confirmation	Repeat same-site nitrate measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: pH	140	Local signal plus direct confirmation	Repeat same-site pH measurement across additional seasons and confirm threshold classifications with a traceable reference workflow.
Kenya, Kwale groundwater samples: TDS	–	Monitoring baseline required	Acquire traceable local TDS records with site, date, unit, method and quality-control fields.
Kenya, Kwale groundwater samples: turbidity	–	Monitoring baseline required	Acquire traceable local turbidity records with site, date, unit, method and quality-control fields.
Malawi, laboratory snapshot: arsenic	10	Data repair required	Recover arsenic units, method, LOD/LOQ and non-detect semantics before threshold analysis.
Malawi, laboratory snapshot: conductivity	31	Data repair required	Recover the conductivity scale/unit convention and raw instrument export; do not repair by assumption.
Malawi, laboratory snapshot: <i>E. coli</i>	–	Organism-specific baseline required	Acquire organism-specific <i>E. coli</i> measurements with sample volume, reference method and response pathway.
Malawi, laboratory snapshot: faecal coliform	19	Organism-specific baseline required	Acquire organism-specific <i>E. coli</i> measurements with sample volume, reference method and response pathway.
Malawi, laboratory snapshot: fluoride	2	Data repair required	Recover the fluoride missing-value codebook and valid observations before threshold analysis.
Malawi, laboratory snapshot: nitrate	31	Local signal plus direct confirmation	Expand repeat-site nitrate coverage and retain method, calibration, operator and quality-control metadata.
Malawi, laboratory snapshot: pH	31	Local signal plus direct confirmation	Expand repeat-site pH coverage and retain method, calibration, operator and quality-control metadata.
Malawi, laboratory snapshot: TDS	31	Local signal plus direct confirmation	Expand repeat-site TDS coverage and retain method, calibration, operator and quality-control metadata.
Malawi, laboratory snapshot: turbidity	22	Local signal plus direct confirmation	Expand repeat-site turbidity coverage and retain method, calibration, operator and quality-control metadata.
Malawi, logger series: conductivity	714	High-frequency sentinel plus chemical confirmation	Pair the two persistently high-conductivity sites and comparison sites with direct chemistry, calibration, fouling, downtime and energy records.

1. Every context also abstains at product level: the framework names no product in any of the 26 contexts, and no row is a national estimate.
2. An en dash means the record count is not available (no accepted measurement population); it does not mean zero exceedances, and absence of records is not evidence of safety.
3. Records are accepted records, except Malawi conductivity, fluoride and arsenic (rows held, which failed acceptance) and the logger series (site-days). Malawi TDS records share the mixed scales that returned data repair for conductivity; the TDS context nonetheless returned local signal plus direct confirmation.
4. The evidence column states what would change an action, not the action it would change to, who supplies the evidence, the sample size, frequency, reference method or cost; for Indian conductivity, TDS and pH it adds a clause not in S2 Table. No reference method is named for a *traceable reference workflow*.
5. *Local labels* are direct measurements used to check a model's scores; a governance review replaces the project conductivity convention with a programme's own action value. LOD/LOQ, limits of detection and quantification. Faecal coliform is never relabelled as *E. coli*.

Decision-relevant cells and record support in India

The Indian archives resolve to 186 state or Union Territory $\times$ parameter cells in 31 states and Union Territories, of which 160 met the support requirement for description; accepted record counts range from 173,029 for conductivity to 9,927 for arsenic (Table 2).

Table 2. Indian monitoring records and decision-relevant cells by parameter. Aggregated from 186 state or Union Territory $\times$ parameter cells; record fractions with Wilson 95% intervals.

Parameter	Accepted records (n)	Records above threshold (n)	Sites (n)	Exceedance threshold	Record fraction [95% Wilson interval]	Described cells (of 31)	Decision-relevant cells (n)	Cells with LOSO support (of 31)
Conductivity	173,029	39,115	24,128	>1,500 $\mu\text{S}/\text{cm}$ (operational convention)	0.2261 [0.2241, 0.2280]	30	1	22
pH	162,729	10,409	23,704	outside 6.5–8.5 pH units	0.0640 [0.0628, 0.0652]	30	4	21
Fluoride	35,843	2,344	32,561	>1.5 mg/L	0.0654 [0.0629, 0.0680]	27	6	14
Nitrate	34,642	6,403	32,206	>50 mg/L as NO_3	0.1848 [0.1808, 0.1890]	27	2	18
TDS	30,151	5,442	14,632	>1,000 mg/L	0.1805 [0.1762, 0.1849]	27	4	14
Arsenic	9,927	31	7,931	>0.010 mg/L	0.0031 [0.0022, 0.0044]	19	0	0

1. Median revisit interval. Annual or longer in all 145 of 145 cells with a recorded median (conductivity 31; pH 31; fluoride 23; nitrate 23; TDS 30; arsenic 7); 48 are exactly 365 days, and the 46 from the Year Book tables cannot fall below the edition spacings (365, 731, 1,096 days).
2. A cell is decision-relevant when its Wilson interval straddles an assumed 0.10 action level: 17 of 160 cells that support description; the classification is not a ranking of severity, prevalence or exposure. Intervals treat records as independent, so the count is a lower bound. LOSO, leave-one-state-out; support for description is not support for generalisation, and for arsenic no cell supports LOSO.
3. Conductivity uses 1,500 $\mu\text{S}/\text{cm}$, an operational project convention, not a health guideline. Nitrate is classified at 50 mg/L as NO_3 ; the transfer analysis uses 10 mg/L as N, the United States maximum contaminant level.
4. Fluoride and nitrate come from the CGWB Year Book tables and the other parameters from the CGWB portal. Records come from all 31 states and Union Territories for conductivity, pH, fluoride, nitrate and TDS; arsenic records come from 23 of them. Per-cell values: S3 Table.

At an assumed action level of 0.10, 17 of the 160 eligible cells are decision-relevant, in 15 states and Union Territories (Fig 5): 6 fluoride cells, 4 pH, 4 TDS, 2 nitrate and 1 conductivity, and none of the 19 eligible arsenic cells, whose record fraction of 0.0031 is not interpretable as a monitoring signal until units, method and detection limits are recovered. These 17 cells are where further records could change a state-level classification; they are not a ranking of severity, prevalence or exposure, and no action in Table 1 depends on them.

Each of the 17 Wilson intervals straddles the action level (Fig 6); the widest, for pH in Nagaland, rests on 44 accepted records and the narrowest, for conductivity in Uttar Pradesh, on 7,280. The count depends on the level, with 29 cells in 16 states and Union Territories at 0.05 and 4 cells in 4 at 0.25. The intervals treat records as independent although sites are revisited, so the counts are lower bounds, and a cell resolved below the action level is not one where monitoring may stop.

The other 143 eligible cells are resolved, with intervals wholly above or wholly below the action level. Rajasthan shows the upper side: of 12,349 conductivity records at 1,641 sites between 2000 and 2024, 6,676 exceeded the 1,500 $\mu\text{S}/\text{cm}$ project convention (54.1%, Wilson 95% interval 53.2% to 54.9%), and of 3,131 TDS records at 1,085 sites,

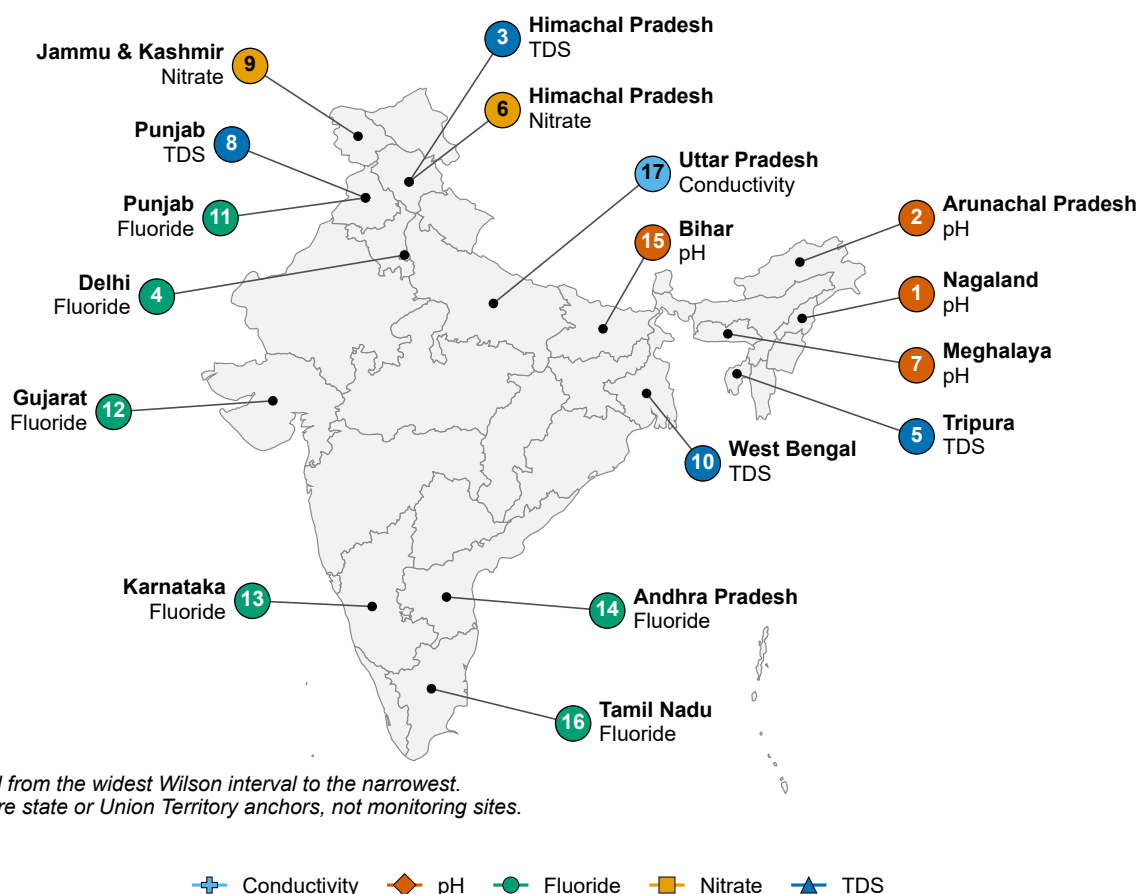


Fig 5. Decision-relevant cells in the Indian archives. Markers locate the 17 state or Union Territory $\times$ parameter cells whose Wilson 95% intervals on the exceedance proportion straddle the assumed 0.10 action level, numbered from the widest interval to the narrowest, an order of uncertainty and not of severity. Markers are administrative anchors, not sampling sites. Boundaries: Natural Earth. Data: S9 Data.

1,627 exceeded 1,000 mg/L (52.0%, 50.2% to 53.7%). Resolved cells such as these call for contaminant-specific confirmation rather than more records, although no Rajasthan conductivity or TDS record states its analytical method.

Record support and revisit cadence set what further analysis the archives can carry (Fig 7). A leave-one-state-out evaluation would be supported in 89 of the 186 cells, including 9 of the 17 decision-relevant cells, and in none of the 31 arsenic cells. In all 145 cells with a recorded median revisit interval the median is annual or longer; 48 are exactly 365 days and the longest is 1,534 days. Because 46 of the 145 medians come from Year Book tables, whose intervals cannot fall below the spacing between editions, and some rest on a single interval, the cadence result describes the archive as held rather than the programme as now designed. No action in Table 1 prescribes a sampling frequency.

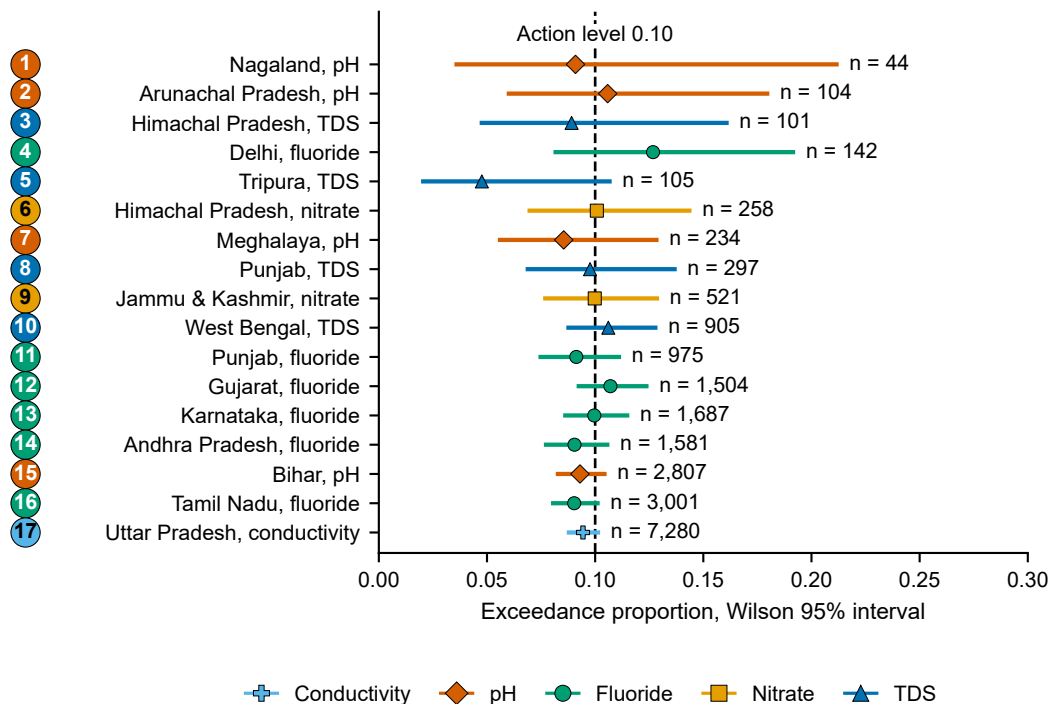
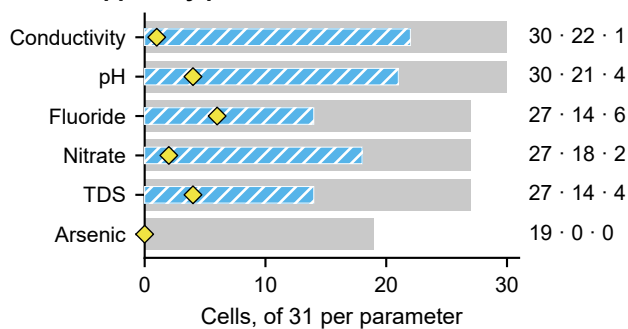
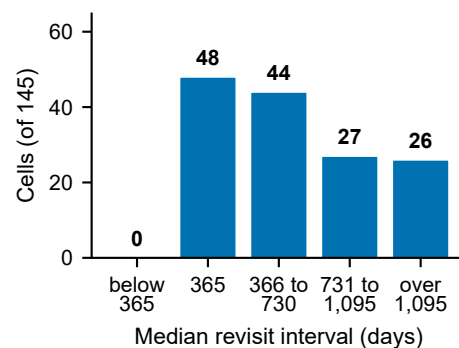


Fig 6. Exceedance proportions in the 17 decision-relevant cells. Points and bars show record proportions with Wilson 95% intervals; n is the number of accepted records. Numbering follows Fig 5 and the dashed line marks the assumed 0.10 action level. The intervals treat records as independent, so 17 is a lower bound. TDS, total dissolved solids. Data: S9 Data.

A Record support by parameter



B Median revisit interval



◆ Decision-relevant ■ Supports description ▨ Supports leave-one-state-out

Fig 7. Record support and revisit cadence in India. (A) Cells per parameter, of 31, supporting description (grey bar), a leave-one-state-out evaluation (hatched bar) and decision-relevant cells (diamond): 160, 89 and 17 of 186 in all. (B) Median revisit interval in the 145 cells with one recorded; none is below 365 days, 48 are exactly 365, and the 46 Year Book cells cannot fall below the spacing between editions. TDS, total dissolved solids. Full values: S3 Table. Data: S9 Data.

Model performance, transfer and the Indian actions

Two model questions bear on Indian sampling: how far a performance estimate depends on validation design, on which no action depends, and whether ordering and probability carry over to archives that took no part in fitting. Validation design changed the performance estimate although the model specification and data were the same, with the model refitted on each design's training partition (Fig 8, Table 3). Across seven tasks from the multinational fitting archive, mean test-partition ROC-AUC was 0.9396 under random-row splitting, 0.8688 under station-grouped splitting and 0.7696 under 3° block splitting. The optimism, random-row minus block ROC-AUC, averaged 0.1701, computed from unrounded means, and ranged from 0.0768 for fluoride to 0.2927 for nitrate. This estimate comes from one partition per design on seven tasks from one predominantly surface-water archive, with no interval.

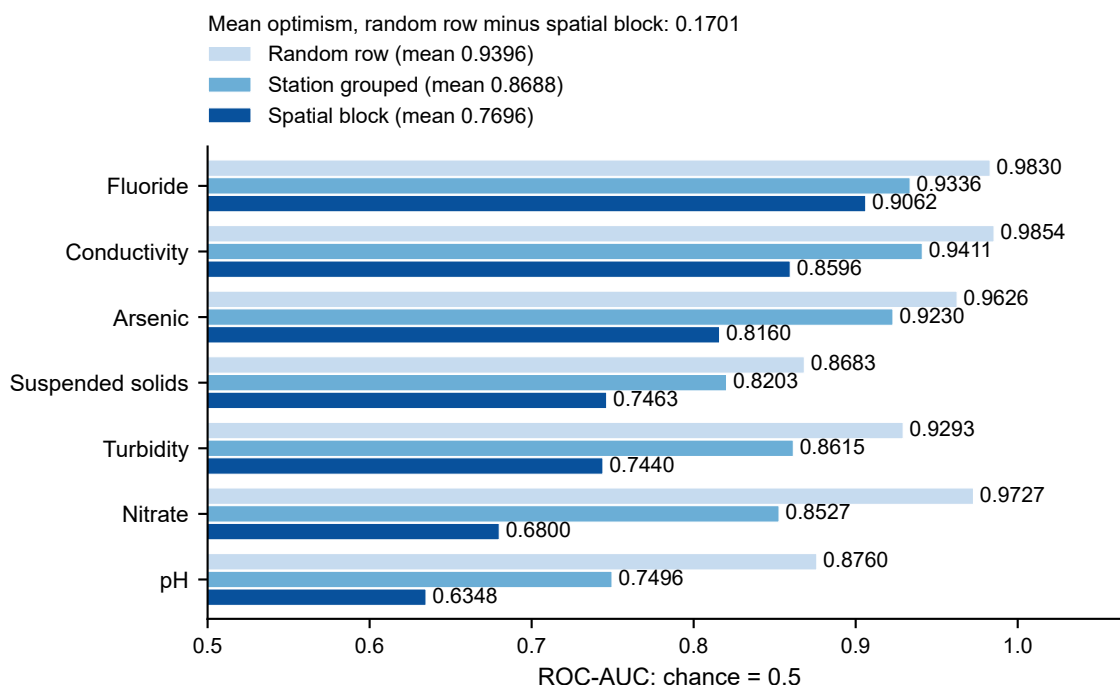


Fig 8. Model discrimination under three validation designs. Test-partition ROC-AUC (area under the receiver operating characteristic curve) for seven tasks under random-row, station-grouped and spatial-block splitting, one partition per design. Optimism is random-row minus spatial-block ROC-AUC, computed from unrounded values. Data: Table 3 and S9 Data.

Under block splitting, precision–recall AUC exceeded each task's overall prevalence for all seven tasks, from 0.0743 for nitrate to 0.9272 for turbidity (Fig 9). Turbidity is the one task in which exceedance is the majority class, with a prevalence of 0.8323; the strict no-skill baseline, the prevalence of the scored test partition, is not reported in the retained tables.

In forward-chained yearly folds, Model B exceeded Model A on each of the six parameters with a paired comparison, by 0.0215 to 0.0643 in ROC-AUC (Fig 10). The advantage needs earlier measurements at the same site, so it is unavailable where monitoring is absent, and it is not evidence of spatial generalisation.

Ordering transferred for five of the 13 archive × parameter pairs evaluated (Fig 11): conductivity and TDS in the CGWB portal, fluoride and nitrate in the CGWB Year Book, and TDS in WQP. TDS was supported in both archives in which it was tested (threshold ROC-AUC 0.7081 in the CGWB portal and 0.7406 in WQP), and pH in neither (0.4870 and 0.4886; the CGWB portal pH interval, 0.4784 to 0.4964, lies wholly below 0.5). Arsenic in the CGWB portal had insufficient support, with one evaluable test on 119 scored observations at 116 sites, and its ordering was inverted: ROC-AUC 0.3170 (95% interval 0.2043 to 0.4450). The nitrate ordering was evaluated at 10 mg/L as N, and its use against 50 mg/L as NO₃ assumes that the ordering does not depend on that difference. Scored observations are fewer than accepted records for every Indian pair; the site exclusion accounts for part of the difference, and the rest is not recorded.

The detected-record severity correlation, which is not a verdict test, had a 95% interval wholly above 0 for 9 of the

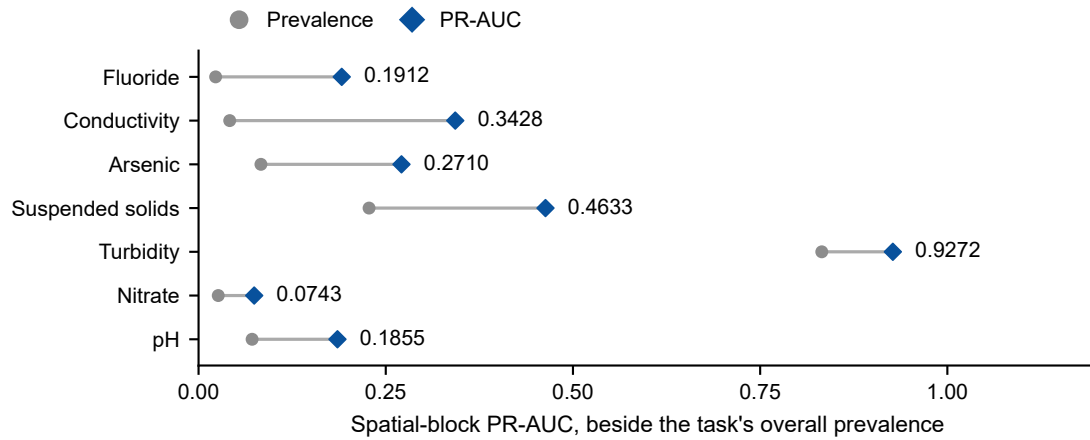


Fig 9. Spatial-block PR-AUC against task prevalence. Precision–recall AUC (PR-AUC) under spatial-block splitting for the seven tasks, beside each task’s overall prevalence, the share of its records above threshold. The test-partition prevalence, the strict no-skill baseline, was not retained. Data: S9 Data.

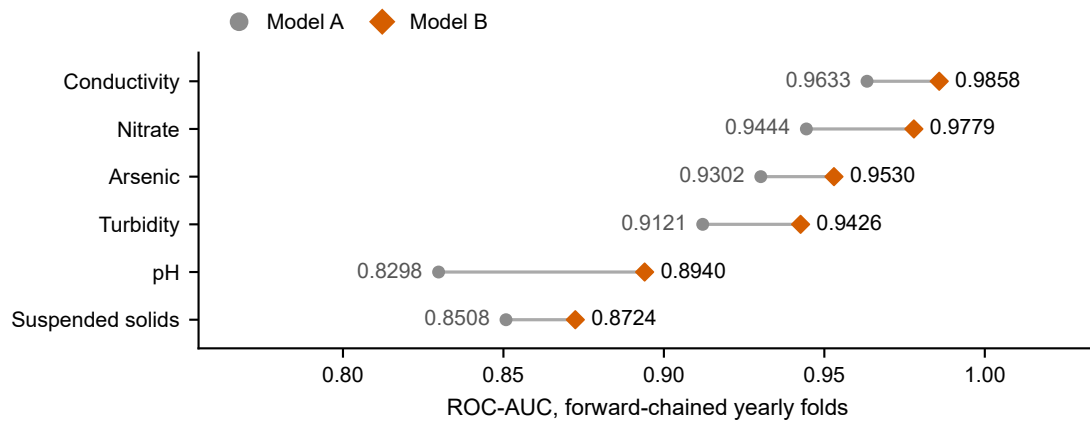


Fig 10. Model A and Model B in forward-chained yearly folds. ROC-AUC (area under the receiver operating characteristic curve) of Model A (environmental, positional and neighbour features) and Model B (earlier measurements at the same site) for the six tasks with a paired comparison, each year predicted from earlier years on records with at least two earlier same-site measurements. Data: S9 Data.

Table 3. Model performance under validation designs and information channels. One row per parameter: test-partition ROC-AUC under three designs, spatial-block PR-AUC and calibration slope, and Model A against Model B in forward-chained yearly folds.

Parameter	Prevalence	ROC-AUC random row	ROC-AUC station grouped	ROC-AUC spatial block	Optimism	Spatial PR- AUC	Spatial calibration slope	Model A ROC- AUC	Model B ROC- AUC	Model B – A
Arsenic	0.0833	0.9626	0.9230	0.8160	0.1466	0.2710	0.8922	0.9302	0.9530	+0.0228
Nitrate	0.0263	0.9727	0.8527	0.6800	0.2927	0.0743	0.8363	0.9444	0.9779	+0.0335
Conductivity	0.0416	0.9854	0.9411	0.8596	0.1258	0.3428	0.7001	0.9633	0.9858	+0.0225
Suspended solids	0.2277	0.8683	0.8203	0.7463	0.1221	0.4633	0.8541	0.8508	0.8724	+0.0215
Turbidity	0.8323	0.9293	0.8615	0.7440	0.1853	0.9272	0.6749	0.9121	0.9426	+0.0305
Fluoride	0.0228	0.9830	0.9336	0.9062	0.0768	0.1912	0.1925	–	–	–
pH	0.0715	0.8760	0.7496	0.6348	0.2412	0.1855	0.2979	0.8298	0.8940	+0.0643
Mean of 7	n/a	0.9396	0.8688	0.7696	0.1701	n/a	n/a	n/a	n/a	n/a

1. Optimism is random-row minus spatial-block ROC-AUC, from unrounded values, so it can differ by 0.0001 from the rounded columns. Random-row values show, by difference, how much record-level splitting inflates the performance estimate; only spatial-block splitting speaks to held-out areas within the sampled geography, and none of the designs is a claim about transfer to an unmonitored country.
2. Model A (33 features, a separate fit from the validation-design columns; S1 Text) and Model B (12 site-history features) were compared in forward-chained yearly folds on records with at least two earlier measurements at the site; Model B exceeded Model A on all six paired parameters in discrimination, and its calibration slope was closer to 1 (all slopes below 1; Model A against Model B: arsenic 0.7608 against 0.8883; nitrate 0.8284 against 0.9344; conductivity 0.7922 against 0.9603; suspended solids 0.8700 against 0.9695; turbidity 0.7689 against 0.9156; pH 0.8442 against 0.9570). The difference under station-grouped or spatial-block splitting was not estimated, and Model B's advantage is unavailable where monitoring is absent.
3. DeLong p-values are not reported: the exported value, 0.00e+00, is below display precision for tests on 174,533 to 803,735 paired observations; the test assumes independent observations, which revisited stations violate, and at these sizes a test reflects sample size, not practical importance.
4. An en dash means not available (fluoride has no paired comparison); 'n/a' means not applicable. No mean is given for the spatial calibration slope (0.1925 to 0.8922) or PR-AUC (0.0743 to 0.9272); read PR-AUC against the prevalence in the same row, a task's base rate rather than population prevalence.
5. Values are from retained result tables (S1 Text).

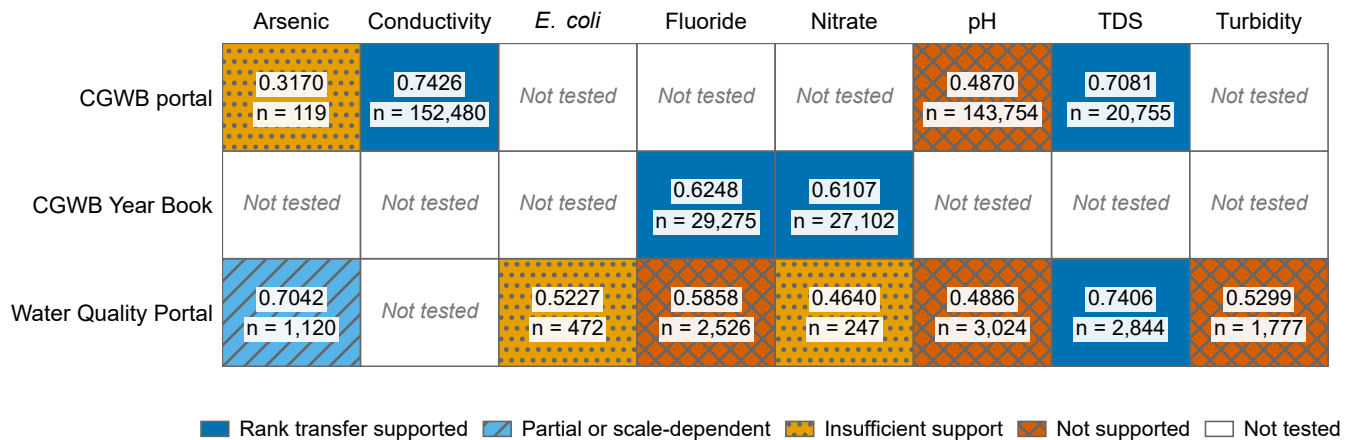


Fig 11. External rank-transfer verdicts by archive and parameter. Cells show the verdict, threshold ROC-AUC and scored observations; 13 of 24 pairs were evaluated. Verdicts follow three interval tests (site-level and within-block rank correlation, and lift in the top predicted decile), not a ROC-AUC cut-off, and insufficient support is kept apart from unsupported transfer. CGWB, Central Ground Water Board; TDS, total dissolved solids. Data: S9 Data.

13 pairs, including four without supported transfer. Among the five supported pairs it ranged from 0.2025 for nitrate in the CGWB Year Book to 0.4778 for TDS in WQP, and their threshold ROC-AUC ranged from 0.6107 to 0.7426 (Fig 12).

Probability did not transfer with ordering (Fig 13). The calibration slope was below 1 on all 13 evaluated pairs, between 0.2383 and 0.4017 on the five pairs whose ordering transferred, and at or below 0 on 4; within the fitting archive the block-design slopes were already below 1 for all seven tasks, between 0.1925 and 0.8922. Brier skill was

● Rank transfer supported ■ Partial or scale-dependent ▲ Insufficient support ✕ Not supported

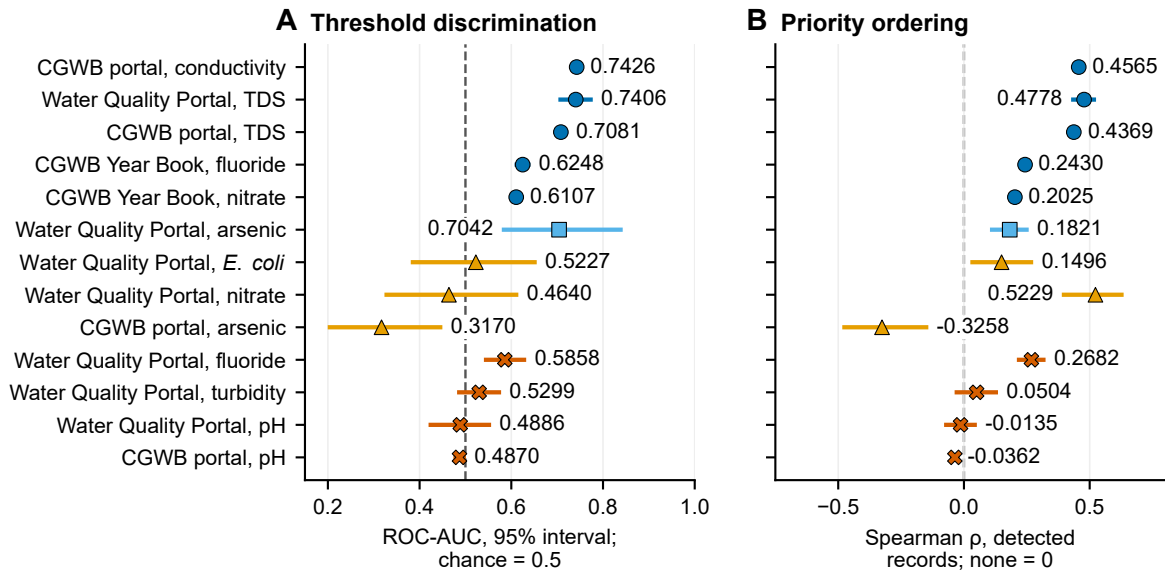


Fig 12. Discrimination and severity ordering under external transfer. (A) Threshold ROC-AUC and (B) detected-record severity correlation (Spearman, predicted risk against observed severity among detected records), each with a site-clustered 95% bootstrap interval (2,000 replicates; 1,998 for Water Quality Portal nitrate), for the 13 evaluated pairs. Severity is $\log_{10}$ concentration, or for pH the distance from the centre of 6.5–8.5; the correlation is not a verdict test. Marker shape and colour give the verdict. Data: S9 Data.

negative on all six Indian pairs and on 11 of the 13 overall; the two non-negative values, 0.0470 for *E. coli* and 0.0169 for arsenic in WQP, belong to pairs without supported rank transfer. The conductivity base rate was 0.0638 in the transfer training data, 0.2261 among all accepted conductivity records and 0.2364 among the scored CGWB portal observations, a shift that any probability carried from the fitting archive would have to absorb.

Calibration-in-the-large was not assessed and local recalibration was not run, so no transferred score can be read as a local exceedance probability. At a fixed 0.5 cut-point most exceeding observations were not flagged; for conductivity in the CGWB portal, sensitivity was 0.0787 (S1 Text gives the counts).

From the transfer verdicts, record support and identity limits, the evidence states the authors assigned returned the Indian actions (Table 1). Conductivity, TDS, nitrate and fluoride returned *rank prioritisation plus direct confirmation*: ordering may set the order in which sources are confirmed, every flagged source still needs a direct measurement, low-ranked sources stay under routine measurement, and no ranking is released with this article. pH returned *direct measurement required*. Arsenic returned *targeted direct reference measurement*, pending method and detection-limit metadata; this study sets no basis for choosing the targets and did not assess laboratory access or cost. Turbidity and *E. coli*, with no accepted Indian measurement population in the archives analysed, returned *monitoring baseline required*; CGWB lists microbial contaminants among additional parameters to be monitored [4], so the evidence may lie with drinking-water agencies rather than in a new survey. Tables 1 and 4 give the evidence that would change each action.

Repeat measurement in Kwale County, Kenya

The Kwale County archive describes groundwater and springs sampled in March and June 2016 and is used without any model. It records 81 sampling locations (Fig 14) across 71 named localities; once surface-water samples are excluded, 73 locations remain, with 139 or 140 records per parameter. Locations are counted as distinct recorded coordinates, so the count depends on how coordinates were recorded. Measured pH lay outside 6.5–8.5 in 41 of 140 records (29.3%, Wilson 95% interval 22.4% to 37.3%), all of them below 6.5, and conductivity exceeded the 1,500 $\mu\text{S}/\text{cm}$ convention in 19 of 140 (13.6%, 8.9% to 20.2%). Nitrate exceeded 50 mg/L as NO_3 in 5 of 139 (3.6%, 1.5% to 8.1%), fluoride exceeded 1.5 mg/L in 3 of 139 (2.2%, 0.7% to 6.2%) and arsenic exceeded 10 $\mu\text{g}/\text{L}$ in 3 of 140 (2.1%, 0.7% to 6.1%). Most locations were sampled twice, so these intervals understate the uncertainty.

● Rank transfer supported ■ Partial or scale-dependent ▲ Insufficient support ✗ Not supported

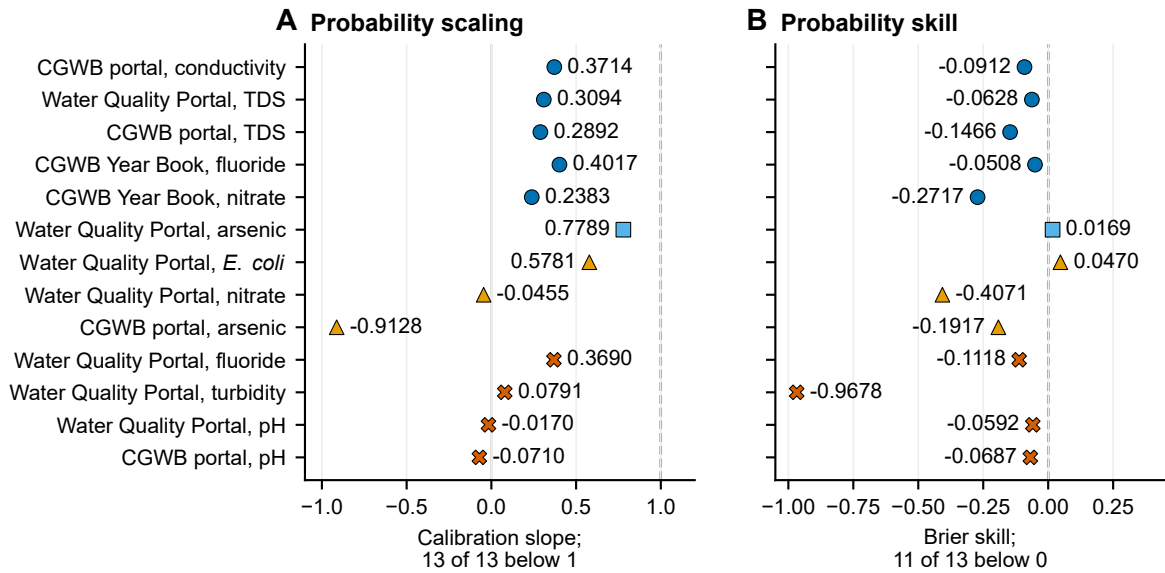


Fig 13. Probability calibration under external transfer. (A) Calibration slope and (B) Brier skill against a reference forecast equal to the external base rate, for the 13 evaluated pairs; reference lines mark slope 1 and skill 0. No local recalibration was run. Marker shape and colour give the verdict; nitrate is scored at 10 mg/L as N. Data: S9 Data.

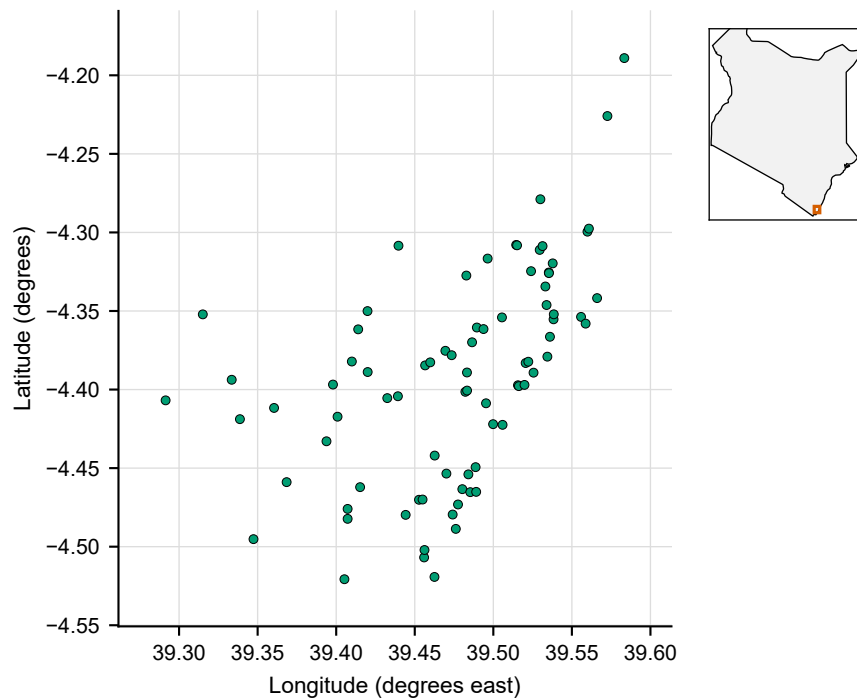


Fig 14. Sampling locations of the Kwale County archive, Kenya. The 81 sampling locations of the Kwale County archive, from two 2016 campaigns [27], reprojected from UTM zone 37S on the Arc 1960 datum without a datum shift; no coastline is drawn at this scale. The text's 73 locations exclude surface water. Inset: Kenya. Openwashdata package under CC BY 4.0. Data: S9 Data.

Two of the three fluoride exceedances, and two of the 19 conductivity exceedances, come from one thermal spring with a recorded temperature of about 59 °C, and the third fluoride exceedance and one June conductivity crossing are unverified values (S1 Text). The counts include these records as the archive holds them.

At the 67 sites sampled in both campaigns (66 with nitrate and fluoride results in both), 15 of the 24 sites with pH outside the range in March were still outside it in June, and 2 more were outside it in June only. Of the 10 sites above the conductivity convention in March, 8 remained above it and 1 more was above it in June only. For nitrate, fluoride and arsenic, one site was above the threshold in both campaigns, and 2, 1 and 1 sites respectively were above it in June only (S11 Table); the fifth nitrate exceedance is at a location sampled once. The source labels the campaigns dry-season and wet-season, but one campaign in each season of a single year does not describe a seasonal cycle.

For pH, conductivity, nitrate, fluoride and arsenic, the rule returned *local signal plus direct confirmation* from the data status alone: repeat measurement at the same sites across further seasons, with threshold classifications confirmed by a reference method the rule does not name (Table 1). Contexts that share the archive and the action could be met by one sampling round. The action sets what to measure next. It does not defer confirmation or a programme's response where a source was recorded above a WHO health-based guideline value in 2016. TDS, turbidity and *E. coli*, for which the archive holds no records, returned *monitoring baseline required*.

Data repair and sentinel monitoring in Malawi

The Malawi laboratory snapshot holds one record per water point for 32 water points sampled between December 2017 and July 2019 (Fig 15). Measured pH lay outside 6.5–8.5 in 7 of 31 records (22.6%, Wilson 95% interval 11.4% to 39.8%), all between 6.24 and 6.42, and the archive flags each as within the Malawian standard it applies. No nitrate record of 31 exceeded 50 mg/L as NO₃, which leaves an upper Wilson bound of 11.0%. Turbidity exceeded the 1 NTU rule in 2 of 22 records (9.1%, 2.5% to 27.8%), both flagged as within the Malawian standard. Faecal coliform was detectable in 4 of 19 samples, all collected in February 2019 and none with a recorded analysis date. Faecal coliform is not *E. coli*, but WHO treats thermotolerant coliforms as an acceptable alternative indicator of faecal pollution [30], so detection at those four water points would ordinarily warrant inspection and resampling whatever the organism-specific evidence gap.

Four parameters could not be read as recorded: conductivity failed a unit-consistency check, TDS is recorded on mixed scales in 10 of 31 records, so no TDS count is reported, fluoride held only a placeholder code, and arsenic had no documented unit. Conductivity, fluoride and arsenic returned *data repair required*, to be made from recovered records and not by assumption; only the originating laboratory or the data publisher can make it, and the rule has no fallback if the records cannot be recovered. The laboratory pH, TDS, nitrate and turbidity contexts returned *local signal plus direct confirmation*. For nitrate, with no exceedance in 31 records, the action amounts to wider coverage; because the TDS records share the mixed scales that returned data repair for conductivity, applying the conductivity rule to TDS would also return data repair. *E. coli*, with no organism-specific measurement, and the faecal coliform context both returned *organism-specific baseline required* (Table 1).

The logger extract holds conductivity from 19 sites in 9 districts, reduced to 714 daily site medians between 31 December 2024 and 7 February 2025, a window of 39 days within Malawi's wet season from November to April [41]. On 76 of the 714 site-days, conductivity exceeded the 1,500 µS/cm convention: two sites, both in Chikwawa District, exceeded it on every recorded day and no other site on any day, so across 695 successive site-day transitions no site changed state. The two sites have median conductivities of about 5,690 and 2,160 µS/cm, and the next highest site median is about 1,360 µS/cm. The series cannot identify the cause: conductivity is not specific to any contaminant, the logger does not record whether values are temperature-compensated, and the extract covers no dry-season days. The rule returned *high-frequency sentinel plus chemical confirmation*, pairing the two persistent sites, and comparison sites, with direct chemistry and with instrument calibration, fouling, downtime and energy records; the analytes, comparison sites, chemistry frequency and logging period are not set by this study.

Screening and testing-service design in Bangladesh and Ghana

In Bangladesh, where no archive was analysed and no model transferred, published kit evaluations bear on whether a field kit can classify a well near the arsenic threshold accurately enough to leave it in service. In a blind evaluation of 123 wells against laboratory measurement, with 65 tested twice, one kit classified 110 wells correctly against the WHO guideline of 10 µg/L and 113 against the Bangladesh standard of 50 µg/L [42]. All 13 of its misclassifications against 10 µg/L were underestimates, in wells between 10 and 27 µg/L [42]. Against 50 µg/L its over- and

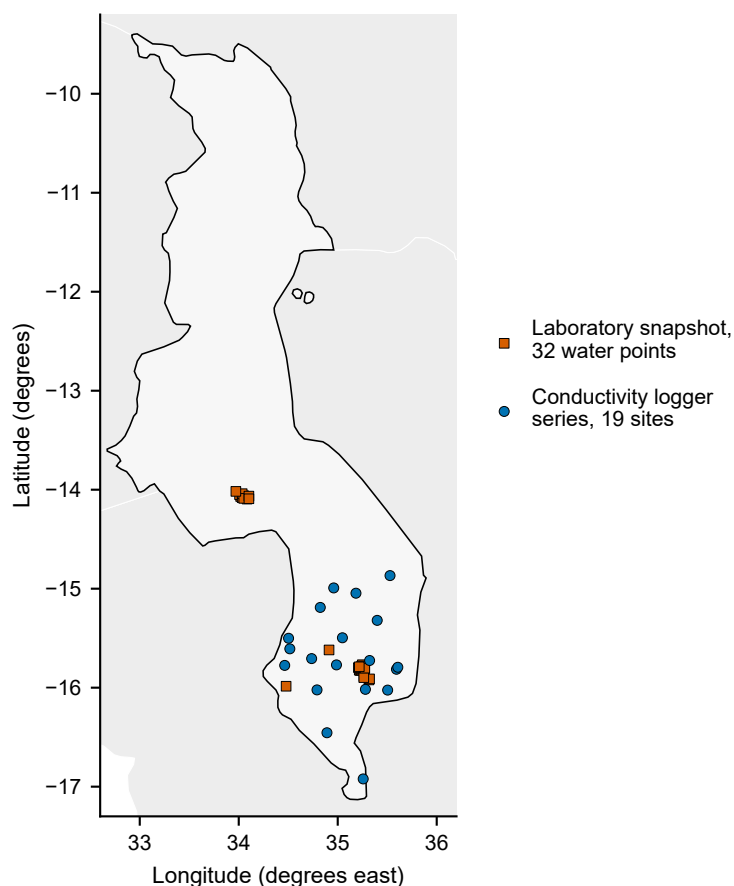


Fig 15. Malawian laboratory water points and conductivity logger sites. The laboratory snapshot of 32 Malawian water points [28] and the conductivity logger series at 19 sites [29]. Nearby markers overlap at this scale. Boundaries: Natural Earth; openwashdata packages under CC BY 4.0. Data: S9 Data.

underestimates were evenly distributed, and the evaluation omitted the kit's hydrogen-sulphide oxidation step [42]. A comparison of eight commercial kits with hydride-generation atomic absorption spectrometry found two accurate and precise in tests that controlled for colour-matching error, four accurate or precise but not both, and two neither [43].

Applied to that evidence, the framework permits screening under a stated decision rule with confirmation of screen positives and of a sample of negatives by a reference method. At 10 µg/L the recorded errors all missed contaminated wells, which is the case for confirming negatives, whereas at 50 µg/L they ran in both directions; the threshold and the fraction of negatives confirmed remain the programme's choice, and kit and laboratory costs in Bangladesh were not assessed. The evidence that would change the action is a false-negative rate near the threshold, with its interval, under the rule a programme uses, together with operator and invalid-test rates.

In Ghana the evidence concerns how to organise *E. coli* testing so that frequency, transport, incubation and staffing remain affordable. Modelled costs per membrane-filtration test for four service approaches, in three study areas in Ghana and Uganda, favoured centralised testing in two of the three [44]. In the Ghanaian study area, centralised testing was cheapest in 96% of simulations, at a median of USD 10 per test [44]. The same study reports the WHO recommendation that samples not be stored for more than 6 h before processing, which links distance to the choice of service, and publishes its cost model for others to run [44]. A review of commercial field tests found none that met the UNICEF target product profile, with the need for a power source the largest limitation and 18 to 24 h of incubation still required [45].

The framework treats the instrument as one component of a service architecture chosen on cost per valid result, which counts invalid and repeat tests; a valid microbial result is one processed within the holding time, incubated as specified and accompanied by passing controls. Until that figure is available, modelled cost per test by service approach, bounded to its study areas and assumptions, is the basis a programme has. Neither costing study states a price year, although one converted costs collected in December 2014 and January 2015 at 1 January 2015 exchange rates [14], and none of the 23 products in the documentary catalogue documents a per-test cost with its basis, currency and date. Neither setting's evidence supports naming a kit or a product.

Technology evidence and monitoring decisions

Wherever an action calls for measurement, the choice of instrument rests on the documentation located for 23 commercial products. The 23-product grid holds them against 54 criteria, 1,242 cells (Fig 16). Of these, 891 (71.7%) are not reported in any located document, 195 (15.7%) are document-sourced, 9 of them from regulatory certification, and 106 were not researched, all for two products. Not reported means that the located documentation is silent: the criterion may still have been measured, and no product has been found wanting. The 9 regulatory cells all belong to one product; they are independent of the manufacturer but are documentation, not device-versus-reference testing. Among the records assembled for the 23 products, paired device-versus-reference field evidence was located only for arsenic, and only one of the three recorded comparisons has recoverable publication provenance.

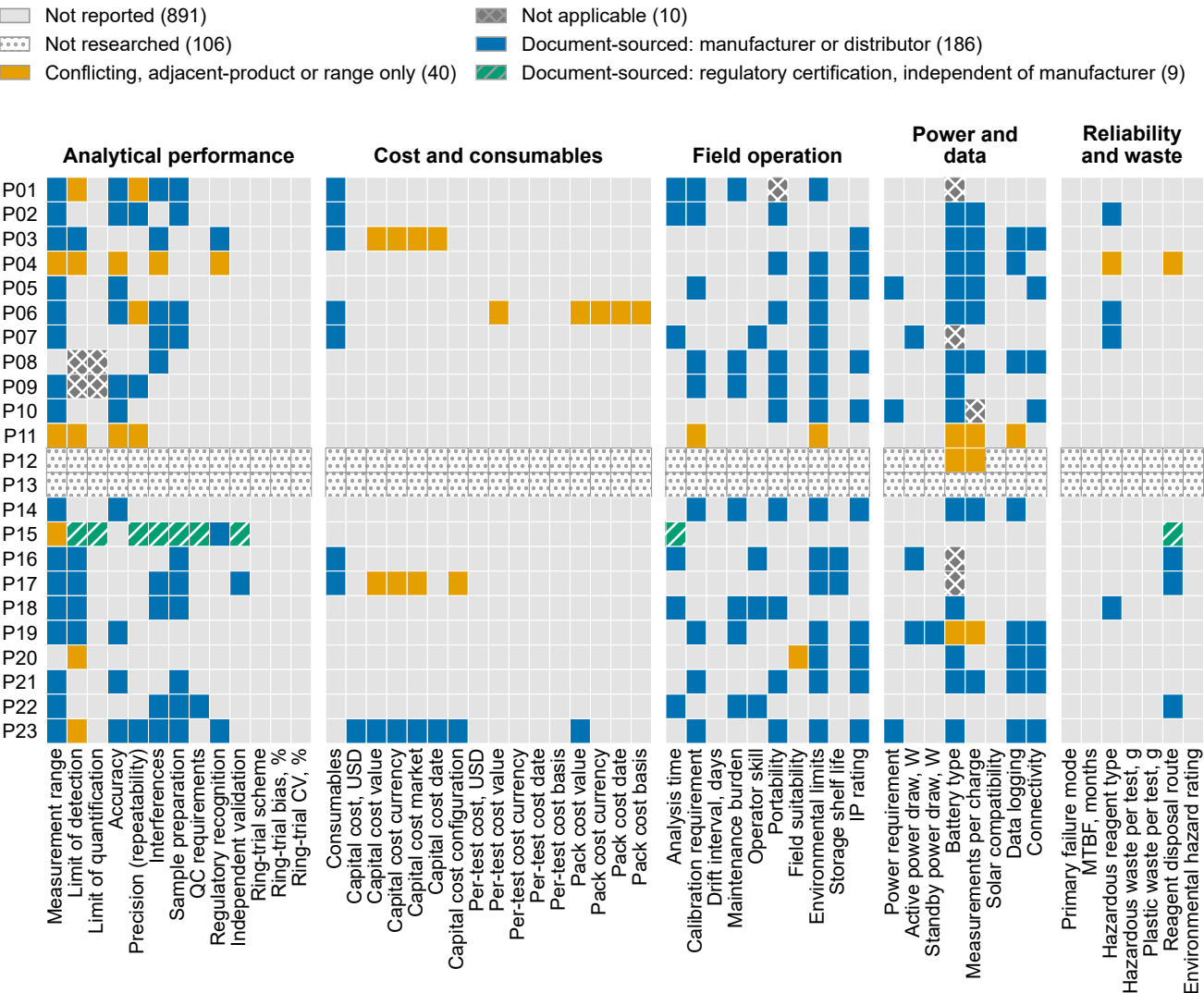


Fig 16. Documentary evidence for 23 products across 54 criteria. Evidence classes of the 1,242 cells: 891 not reported in any located document (unknown evidence), 106 not researched, 40 conflicting, adjacent-product or range-only, 10 not applicable and 195 document-sourced, 9 from regulatory certification. Identities and sources: S4 Table. MTBF, mean time between failures; QC, quality control; CV, coefficient of variation; IP, ingress protection. Data: S9 Data.

Under the two counting rules set out in Methods, 1 of the 11 active criteria meets the documentation threshold on the strict field and 4 on any admissible field (Fig 17). None of the 23 products, including each of the 21 that were researched, documents all 11 active criteria. On any admissible field the most any product documents is 9, reached by 1 product, with 4 products documenting 8 and 4 documenting 7; on the strict field the most is 7. The outcome therefore rests on requiring all 11: a rule requiring 9 would admit one product. For 19 of the 54 criteria, no located

document covers any product. In all 26 contexts the framework named no product, from one determination over the 23-product catalogue that is not evidence about the contexts.

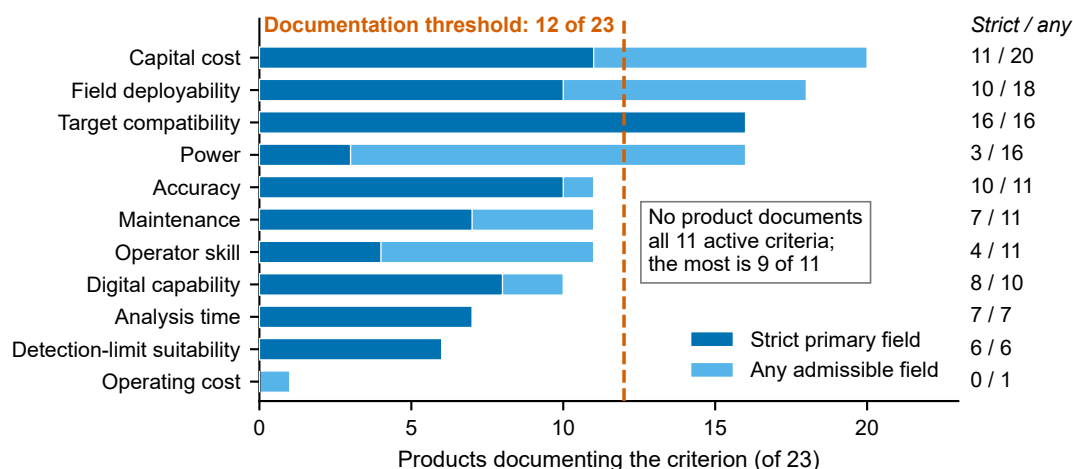


Fig 17. Coverage of the 11 active criteria. Products documenting each active criterion on strict primary fields and on any admissible field, against the documentation threshold of 12 of 23 products (dashed line). No product documents all 11 active criteria; the most any documents is 9. Data: S9 Data.

Among the ten technology archetypes, a population separate from the named products, one archetype led under every cost mapping in 5 of 7 weighting scenarios, and in the other two the leader changed with the cost mapping (S7 Table). A leading archetype is preference context under stated weights, not a product or the class of instrument to use.

Table 4 brings the settings together: for each, the evidence held, what a model may and may not do, the action returned and the evidence that would change it; S12 Table adds the monitoring decision each setting poses and where the framework declines.

Table 4. Monitoring evidence and actions across application settings. One row per setting; Bangladesh and Ghana rest on published evidence only, and Table 1 lists the 26 contexts behind the India, Kenya and Malawi rows.

Setting	Evidence held	What a model may do	Action returned	Evidence that would change it
India (CGWB portal and CGWB Year Book)	186 cells in 31 states and Union Territories; 160 eligible; 17 decision-relevant at a 0.10 action level, which locate where further records could change a classification and set no action; median revisit interval annual or longer in all 145 cells with one recorded	Ordering may sequence confirmatory sampling for conductivity and TDS (CGWB portal) and fluoride and nitrate (CGWB Year Book); pH not supported; arsenic insufficient support; no pair licenses a probability	Rank prioritisation plus direct confirmation (4); Monitoring baseline required (2); Direct measurement required (1); Targeted direct reference measurement (1)	Local direct measurements permitting recalibration; persistent well, date and sample identity; arsenic method and detection-limit metadata; accepted turbidity and <i>E. coli</i> populations
Kenya, Kwale County (groundwater and springs)	Records above threshold: pH 41 of 140, conductivity 19 of 140, nitrate 5 of 139, fluoride 3 of 139, arsenic 3 of 140; 67 sites sampled in both 2016 campaigns	None; no model was fitted or scored	Local signal plus direct confirmation (5); Monitoring baseline required (3)	Repeat same-site measurement across further seasons of the year; 9 of 24 sites outside the pH range in March were inside it in June
Malawi, laboratory snapshot	Records above threshold: pH 7 of 31, nitrate 0 of 31, turbidity 2 of 22; faecal coliform detectable in 4 of 19; conductivity, fluoride and arsenic not readable as recorded, and TDS on the same mixed scale as conductivity in 10 of 31 records, so no TDS count is given	None; no model was fitted or scored	Local signal plus direct confirmation (4); Data repair required (3); Organism-specific baseline required (2)	Recovered conductivity and TDS scale, fluoride codebook and arsenic units; organism-specific <i>E. coli</i> measurement; wider repeat-site coverage

Setting	Evidence held	What a model may do	Action returned	Evidence that would change it
Malawi, conductivity logger series	19 sites, 714 site-days over 39 days; 76 site-days above 1,500 $\mu\text{S}/\text{cm}$; 2 sites above on every recorded day; 0 changes of state in 695 transitions	None; temporal description only	High-frequency sentinel plus chemical confirmation (1)	Direct chemistry with calibration, fouling, downtime and energy records at the persistent and comparison sites
Bangladesh (published evidence only)	One kit classified 110 of 123 wells correctly against 10 $\mu\text{g}/\text{L}$ and 113 against the Bangladesh standard of 50 $\mu\text{g}/\text{L}$; all 13 errors at 10 $\mu\text{g}/\text{L}$ were underestimates, and at 50 $\mu\text{g}/\text{L}$ errors ran in both directions; in tests controlling colour-matching error, two of eight kits were accurate and precise	None; no model transferred	Screening under a stated rule, with reference confirmation of positives and of a sample of negatives; the threshold and the fraction confirmed are the programme's to set	A false-negative rate near the threshold, with its interval, under the programme's rule; operator and invalid-test rates
Ghana (published evidence only)	Modelled cost per test favoured centralised testing in two of three study areas in Ghana and Uganda; in the Ghanaian area it was cheapest in 96% of simulations; no reviewed field test met the UNICEF target product profile	None; no model transferred	Choose a service architecture on cost per valid result, with the instrument as one component	Network-specific cost per valid result, including invalid and repeat tests, and the sampling frequency at which onsite testing becomes cheaper

1. Counts above a threshold are record fractions within each archive, not prevalence, and a state or Union Territory cell is not a well or a person; thresholds as in Methods (arsenic 0.010 mg/L is 10 $\mu\text{g}/\text{L}$).
2. Kwale County and the two Malawi archives are bounded local archives, not national samples; the Kwale counts exclude surface water, and two of its three fluoride exceedances come from one thermal spring.
3. Bangladesh and Ghana had no archive analysed and no model transferred; the kit counts are for one kit at the two thresholds stated, and the costs are modelled.
4. 1,500 $\mu\text{S}/\text{cm}$ is an operational project convention, not a health guideline. Ordering that transfers licenses the sequencing of confirmatory sampling, never a probability.
5. No product is named in any setting. S12 Table adds each setting's monitoring decision and where the framework declines.

Discussion

Implications for monitoring practice

Taken together, the applications show which kind of evidence carried each action. The 18 Kenyan and Malawian actions came from record harmonisation and data status alone, and in Malawi harmonisation turned unreadable parameters into a defined data-repair task rather than a silent gap. The Indian actions drew on record support and on the transfer verdicts, which permitted ordering in four of the six Indian pairs and supported no probability. The validation designs entered no action; with one partition per design on seven tasks, the fall in mean ROC-AUC from random-row to block splitting illustrates why the framework ties a model's permitted use to the design rather than to the metric. The documentary audit turned missing product information into a criterion-by-criterion account of what no located document reports (S4 Table).

For India, the results permit ordering to sequence confirmation within the four supported archive $\times$ parameter pairs, rather than supporting a hazard map. Where ordering transferred, it could decide which conductivity, TDS, nitrate or fluoride sources to confirm first, though not which may be left untested. At the assumed 0.10 action level, the 17 decision-relevant cells locate where further records could change a state-level classification. Because calibration slopes were below 1 on every transfer pair and no pair was shown to be adequately calibrated, the framework withholds probability-based use; because the transfer also crossed water-body type, the supported pairs do not separate a change of geography from a change of water type. These are implications of retrospective results, not a tested operational gain.

In Kwale County and Malawi the record itself set the next step. In Kwale that step is returning to the same sources, since pH classifications changed between campaigns at 11 of the 67 sites sampled twice, and the fluoride signal rests on a thermal spring and an unverified value that a practitioner would check before acting. In the Malawi laboratory snapshot, four parameters could not be read as recorded, and three of them returned data repair, which needs units, a missing-value codebook and, for arsenic, detection limits to be recovered; these gaps describe the records and say nothing about the water. The snapshot's gaps are in line with reports that groundwater monitoring

data in Malawi are persistently scarce, a scarcity made worse by existing monitoring networks that are not operating [46]. The logger series shows persistence at two sites without identifying its cause. Both sites are in Chikwawa District, and groundwater salinity too high for potable use has been documented in Malawi's lower Shire Valley [41]; whether that explains these sites was not assessed.

In Bangladesh and Ghana the constraint lay in the decision rule and the service design rather than in any model. In hypothetical well-switching scenarios for Araihaazar, Bangladesh, relying on kit rather than laboratory data lost only a modest part of the reduction in average arsenic exposure, set against the logistical and financial challenge of supplying laboratory data alone [47]. The same analysis found the 50 µg/L threshold used in Bangladesh, rather than the WHO guideline of 10 µg/L, close to optimal for average exposure reduction [47]. A programme could accept kit screening on that basis while setting the residual error it will tolerate. For Ghana, both costing studies cited here, one across sub-Saharan Africa and one covering study areas in Ghana and Uganda, express cost per test with quality-control samples added as a markup [14,44], whereas a service design needs cost per valid result, counting invalid and repeat tests.

Implications for model use and technology assessment

Mean discrimination fell from 0.9396 to 0.7696 between two validation designs applied to the same models and data, whereas a review of machine-learning models for United States drinking water found tenfold cross-validation or a randomly chosen test set the most common basis for reported performance [18]. In a forest biomass mapping study, non-spatial validation suggested that a model predicted more than half of the variation, whereas spatial validation found almost no predictive power [48]. A methodological source on spatial dependence records the counter-argument that spatial cross-validation is not the right way to evaluate map accuracy [9], so the single spatial-block split is used here to address a generalisation claim, not to certify accuracy away from sampled locations. Whether the national hazard models for India cited in the Introduction carry such optimism was not tested here. What the framework contributes here is a rule that ties the validation design to the use a model is allowed, so that a random-row figure cannot authorise decisions in unsampled places.

The calibration result has precedents. Estimated risks can be unreliable even when discrimination is good, for example overestimating risk when a model is used in a setting where incidence is lower [11]. Calibration-in-the-large, the calibration slope and discrimination have been proposed as distinct measures of model performance, with problems of calibration-in-the-large often found at external validation [49], and moderate calibration has been proved to guarantee decision-making that does no harm [50]. No transferred score here was shown to be moderately calibrated, so none carries that guarantee; the slopes show that transferred probabilities were miscalibrated, not every form the miscalibration takes. The permission gate turns this distinction into a decision rule: a model whose ordering transfers may sequence confirmatory sampling, and only local calibration can license a probability.

Evidence from outside the product catalogue points the same way as the documentary shortfall. Besides the operator-dependent kit verification cited in the Introduction, a laboratory evaluation of a marketed low-cost sensor found no correlation between its score and the presence of *E. coli* [51]. Both evaluated one product at a stated date, and no conclusion about any product is drawn from them. Most of 72 institutions assessed across ten sub-Saharan African countries did not achieve the microbial testing levels set by applicable standards or WHO guidelines [52], and drinking-water surveillance often falls short of its required frequency [2]. Threshold-near performance, operator dependence and achievable testing frequency are what a programme would need documented, by suppliers, service records or independent evaluation, before it could rely on a named instrument; none is among the active criteria audited here.

Abstention here differs from its usual machine-learning form, in which a model abstains where it is likely to make a mistake [17]. The framework abstains where the evidence needed for a decision is absent, so its abstention describes the evidence base rather than predictive error. Stochastic multicriteria acceptability analysis aids decisions in problems with inaccurate, uncertain or missing information [40], and the archetype comparison draws on it, but no sampling of weights can supply a criterion value that no located document holds. We therefore read the documentary audit as the substantive result, and the product-level abstention as the output of the sufficiency rule applied to it; the audit shows which documentation is missing before named products can be compared.

Applying the framework in other settings

Applying the framework elsewhere would need at least the following steps, several of which go beyond what was done here. First, identify and quality-check the local archives and their provenance. Second, align units, analytes and source types, and justify thresholds locally. Third, assess site identity, record support and spatial and temporal dependence before choosing validation designs. Fourth, validate any model's ordering in the new setting and calibrate it on local measurements before its probabilities are used. Fifth, assemble technology and deployment evidence, including threshold-near performance, invalid-test rates and service constraints. Sixth, assess the feasibility of the returned actions with practitioners and test prospectively whether they change monitoring decisions for the better.

Two observations would test the framework itself. It would gain support if the same rules, applied to a catalogue with fuller documentation, admitted a named product, since that would show that the abstention reported here reflects this evidence base rather than rules no product could meet. It would be shown to be underspecified if two analysts applying the stated rules to the same archive returned different actions. No one outside this project has yet applied the rules, the framework has not been compared with an alternative approach, and no monitoring programme has reviewed the returned actions for feasibility or cost; the applications reported here are retrospective.

The evidence that would change the actions reported here is specific: local measurements that allow model recalibration, persistent site and well identity, paired chemistry at sentinel sites, threshold-near kit performance under a stated rule, and invalid and repeat tests recorded in a deployed network, from which cost per valid result follows. S8 Table sets out these five and 11 further research priorities in four families. The measurements a recalibration needs are themselves direct measurements, so its value to a programme depends on the model then reducing later measurement, which was not assessed.

Conclusion

The evidence-to-action framework developed here links archived records, model outputs and instrument documentation to specific monitoring actions, each stated with the evidence required to reassess it and recorded with the rule and evidence it rests on. Applied retrospectively to 26 monitoring contexts in India, Kwale County in Kenya and Malawi, it returned eight distinct actions, and its decision logic was applied to published evidence from Bangladesh and Ghana. Where only records were available, as in the local archives of Kwale and Malawi, their state set repeat measurement, baselines, data repair and paired chemistry; in India, at an assumed 0.10 action level, they located at least 17 of 160 eligible cells where further records could change a classification. Where models were available, permission came from transfer tests rather than a headline metric: ordering that can sequence confirmatory sampling within those archives was supported for four of six Indian archive $\times$ parameter pairs, and no transferred score was used as a probability. The fall in mean ROC-AUC from 0.9396 under random-row to 0.7696 under spatial-block splitting, on one partition per design, illustrates why a model's permitted use follows its validation design. Where an instrument was to be chosen, no product documented all 11 active criteria in any located document, so the audit shows which documentation is missing; for Bangladesh and Ghana the same logic framed screening with confirmation and a service design chosen on cost per valid result, a quantity neither costing study reports.

For another country the route is conditional on its evidence. Where accepted local records exist, with a stated chemical basis and locally justified thresholds, the rules can return confirmation, repeat-measurement or repair actions without a model. Where a model is to guide sampling, its ordering must first hold on local measurements, and its probabilities need local calibration before they are compared with an action level. Where an instrument is to be chosen, supplier documentation, service costs and practitioner review must cover the criteria the programme requires. No one outside this project has yet applied the rules, and the next test is prospective: whether the returned actions improve confirmatory sampling under local operating constraints.

Acknowledgments

We acknowledge the originators of the water-quality records held in GEMStat and the UNEP GEMS/Water Programme, whose database supplied the records used to fit the models. Indian groundwater-quality data were produced by the Central Ground Water Board and obtained through the National Water Data Portal of the National Water Informatics Centre, Ministry of Jal Shakti, and from the Board's Ground Water Quality Data documents. United States records were obtained from the Water Quality Portal of the National Water Quality Monitoring

Council, the U.S. Geological Survey and the U.S. Environmental Protection Agency. The Kenyan and Malawian local archives are openwashdata packages. Soil properties are SoilGrids 2.0 data obtained through ISRIC – World Soil Information; land cover is ESA WorldCover 10 m 2021 v200; and population density is from WorldPop.

Further acknowledgments: TO BE FILLED LATER

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Supporting information

S1 Text. Harmonisation, model configuration and decision rules. Data rules and accepted populations, model and transfer configurations, decision-rule branches, source documents and the scope of result verification.

S2 Table. The 26 monitoring contexts in full. Every field of the context matrix and the definitions of the eight actions.

S3 Table. Support, cadence and exceedance for 186 Indian cells. Per-cell counts, Wilson intervals and revisit intervals behind Table 2, Fig 5, Fig 6 and Fig 7.

S4 Table. The 23×54 product documentation grid. Evidence class, value, source document and locus for each cell, with product identities (P01–P23 in Fig 16).

S5 Text. Mechanistic ablation. Rung definitions, configuration, layer sources and discrimination with block-bootstrap intervals.

S6 Fig. An illustrative conductivity prediction surface. One held-out window, read as an ordering and not a probability; its design counts and layers are given with the figure.

S7 Table. Rank acceptability of technology archetypes in seven weighting scenarios. Rank probabilities across cost mappings; archetypes are not products.

S8 Table. Sixteen research priorities in four families. Study design, endpoint and the decision each priority would inform.

S9 Data. The values plotted in the main-text figures and S6 Fig. One CSV file per figure.

S10 Text. Record of artificial-intelligence assistance. Each activity, its inputs and outputs, and its verification.

S11 Table. Record fractions and campaign comparisons for the Kenyan and Malawian archives. Wilson intervals, the Kwale paired campaigns and the Malawi logger summary; the archives are local, not national samples.

S12 Table. The setting synthesis in full. All seven columns of the setting synthesis, five of which appear in Table 4, including each setting's monitoring decision and where the framework declines.